The decision clock: review intervals as a design lever for stable resource governance

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Abstract

Fisheries are managed on a schedule: stocks are assessed at fixed intervals and the resulting catch limits held until the next review. Standard models treat this institutional response as a continuous delay or a single annual step. We show the substitution matters. Modelling periodic review as sample-and-hold governance — a loop that observes and updates only at review times — we establish two exact properties, forward invariance of the sampled state space and rapid-review consistency over finite horizons only, and compute stability review-interval by review-interval. Stability boundaries move or vanish when the operator is changed: at an illustrative baseline the exact map crosses once near a 6.5-year interval while one-step approximations report artefact crossings, and the protective channel is stable at every tested interval. A multiplicity-controlled screen of 42 stocks finds no robust institutional cycles and a 32-system case search no unconfounded oscillator, showing periodicity alone cannot diagnose governance feedback; the northern cod record splits into a formulation-dependent crash and an unresolved post-collapse identification problem. What decides whether management stabilises or destabilises a fishery is therefore not biology alone but the decision clock — the timing and form of policy revision — a design variable with stability consequences, to be treated explicitly in management strategy evaluation.

Keywords: periodic review; sample-and-hold governance; review interval; institutional delay; sampled-data control; Neimark–Sacker bifurcation; fisheries management; decision clock; management strategy evaluation

Evidential status. Every central claim carries its exact status — exact, conditional, archived, provisional, or prospective — in the statement inventory (Supplementary S1); the main text states each result once with its record pointer, and Section 4.7 collects the limitations.

1 Introduction

Fisheries governance is periodic. Surveys are scheduled, assessments are produced at fixed cadences, advice is delivered at review points, and the resulting catch or effort controls are held in force until the next review. The review interval — the time between successive opportunities to revise the command — is sometimes a single year, sometimes a multi-year plan, sometimes a moratorium of indefinite length (Punt and Donovan, 2007; DFO, 2016). The formal literature treats this institutional clock in two ways, neither of which matches the operating object. Population-dynamic theory models institutional response either as a continuous-time lag, that is, a delayed signal entering an evolving control law, in the lineage of Gurney, Blythe, and Nisbet (1980); or as an annual discrete-time decision appended to a surplus-production model, in which the between-decision dynamics are compressed into a single annual step. Real institutions do neither. They observe and assess at discrete times, choose a rule-based command,

and then hold or interpolate that command until the next decision opportunity. This paper calls that architecture *sampled governance* — a governance loop in which state observation and control update occur at discrete review times, not continuously. It is the same architecture the title and abstract call *sample-and-hold governance*: one object under two names — *sampled governance* names the institutional loop, *sample-and-hold* its control-theoretic update law — and the article body uses *sample-and-hold* (with *sampled* for its maps and loops) throughout.

Two failure modes follow from this mismatch. The first is *architecture substitution*. Replacing the sample-and-hold architecture by a continuous delay, or by an annual difference equation, can move or delete stability boundaries. Conclusions about a governance system then depend on an operator that the institution does not implement. The second is *causal promotion*. Spectral peaks, cohort cycles, and crisis-driven monitoring records circulate as evidence for institutional-feedback mechanisms when they carry no information about the controller's sign, cadence, or timing at all. The marine-science literature already contains the disciplinary template for resisting this promotion. The skeptical re-analysis of the Russell Cycle (McManus, Licandro, and Coombs, 2016) showed that an 88-year zooplankton series long treated as a cycle failed to show a resolvable true periodicity. This article extends that discipline from climate-forced cycles to governance-feedback claims.

The contribution is fivefold. First, a sample-and-hold model of periodic review is specified, in which the governance clock is decomposed into seven time objects (observation interval, assessment interval, review interval, decision lag, deployment lag, ecological response lag, and memory timescale) and the extractive controller is an explicit discretisation with specified comparator classes. Second, two properties of the sampled process are established exactly: forward invariance of the sampled state space, and the precise scope of the rapid-review limit — a finite-horizon numerical-consistency statement that implies nothing about stability at positive review intervals. Third, the review-interval spectrum is computed under the discipline that the sample-and-hold map and the continuous-delay equation are *different operators*. A stability window located on one map does not transfer to the other, and the paper states which operator carries which statement. Fourth, an empirical layer is reported: a multiplicity-controlled spectral screen of 42 annually assessed stocks, a power analysis of that screen, and a structured case search across more than thirty resource systems. Fifth, the northern cod case (NAFO 2J3KL) is carried through the resulting falsification discipline. These five are retrospective; the paper's constructive content is the prospective programme — five prospective designs specified as preregistration targets — that could convert or refute the mechanism.

The model class examined here is the extractive controller: a rule that raises extraction effort in response to a perceived decline. This is the demand-side form of the overshoot loop. The response to a perceived shortfall falls on the productive stock itself rather than on its regeneration. Extraction is raised precisely because the base has declined, which accelerates the decline it is reacting to. Its protective counterpart (effort reduced in response to decline), fixed multi-year plans, and hybrid rules serve as comparators.

The extractive rule is not proposed as desirable management. It is a diagnostic representation of perverse pressure: when a perceived shortfall threatens short-term employment or revenue, the political-economic response — subsidies that sustain effort, pressure to hold catches up — can push extraction upward as the base declines. Capacity-enhancing subsidies are the documented form of this pressure (Sumaila et al., 2019). The protective controller is the policy-relevant comparator; the pair exists to show how controller sign interacts with review timing, not to recommend the extractive rule.

Conclusions drawn inside the extractive class are not generalised to governance as a whole, and the screen below asks whether such a channel is empirically resolvable at all. The implication for institutional design is direct: an annual-review extractive rule and a multi-year protective plan are not interchangeable instruments, and a sampled-data analysis must state which operator a given stability claim is computed on.

The architecture is not unique to fisheries. National climate pledges under the Paris Agreement are communicated every five years and informed by five-yearly global stocktakes (UNFCCC, 2015, Articles 4.9 and 14): held commands under periodic review (Figure 1A). Failures then take a common form across sectors: the governance clock (the review interval), the ecological clock (the recovery rate), and the environmental clock (exogenous cycles such as ENSO, Section 3.7) fall out of phase. The timing and form of policy revision — the review interval together with the revision rule — is the decision clock. This article develops the fisheries case in full; Section 4.8 returns to the cross-sector reading.

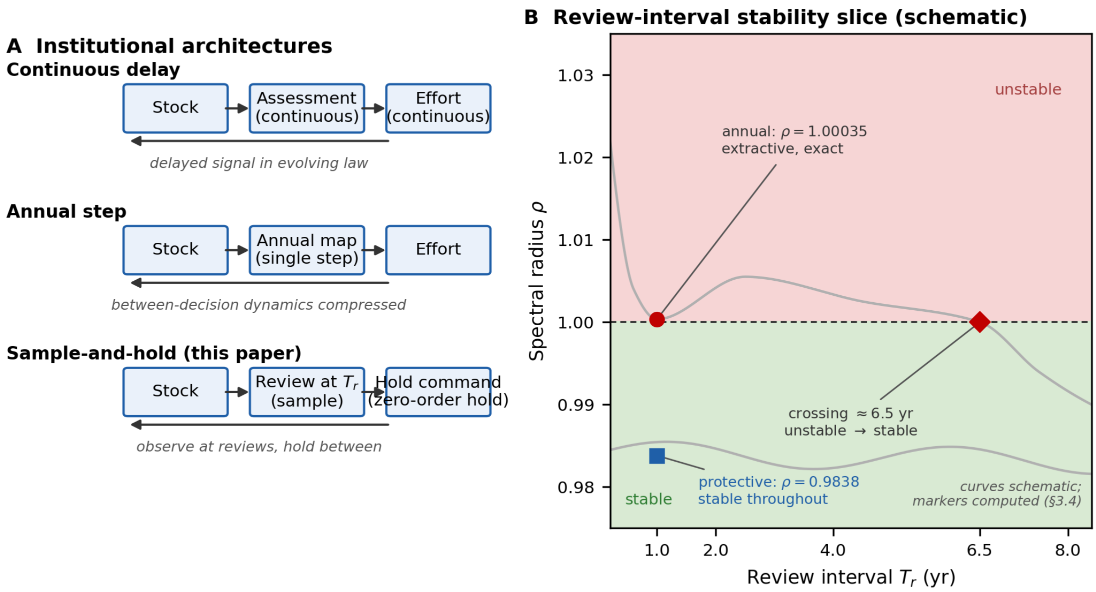

Figure 1: The decision clock. **A:** Three institutional architectures: continuous-delay feedback, the annual-step compression, and sample-and-hold governance, which observes at review times and holds the command between them. **B:** Review-interval stability slice (schematic): the extractive exact update is marginally unstable at annual review ($\rho = 1.00035$) and crosses once near a 6.5-year interval (unstable → stable), while the protective update is stable throughout ($\rho = 0.9838$ at annual review). Curves are schematic; markers show computed values (Section 3.4).

2 Material and methods

2.1 The sample-and-hold model of periodic review

The model treats periodic review as a sample-and-hold control architecture: the command set at a review instant is held fixed until the next review, and only the between-review ecological dynamics are continuous in time. Seven time objects are kept distinct, and no two are collapsed into a single "governance lag" unless the empirical record cannot resolve them and the aggregation rule is stated (Table 1).

For fisheries readers, the control-theoretic vocabulary maps onto management-procedure language: the controller is the harvest control rule; the command is the catch or effort limit it sets; the review interval is the advice cycle; the hold is the plan period between revisions; and the operator — the decision architecture on which a stability claim is computed — is the management procedure itself. Assessment error and implementation error enter where observation meets advice and where decisions meet deployment (Table 1).

Table 1. The governance-time ontology.

Object	Definition
Observation interval T_{obs}	Time between raw measurements
Assessment interval T_{assess}	Time between formal state estimates
Review interval T_r	Time between opportunities to change the command
Decision lag τ_{dec}	Assessment completion to formal decision
Deployment lag τ_{dep}	Decision to implemented change in extraction pressure
Ecological response lag τ_{eco}	Implementation to detectable ecological response
Memory timescale τ_m	Relaxation time of a filtered institutional signal; not a discrete delay

observation → assessment → advice → decision → implementation → ecological response

The chain is the sequence Table 1 resolves into distinct objects; collapsing any two stages without a stated aggregation rule is the measurement-level form of architecture substitution.

Reviews can be annual while deployment is delayed; observations can be frequent while decisions are legally fixed for several years; and stock and realised-effort series alone may identify only a combined closed-loop phase shift rather than the separate τ_{dec} and τ_{dep}. Separating them empirically requires dated observation releases, assessment products, commands, and implementation records. Alternatively, a proved structural-identifiability argument for the specified observation model can serve. Two further conventions apply. The review opportunity is not the same as the response sign: the extractive command studied here raises effort in response to a perceived decline, and the protective controller enters only as a comparator. The fixed-τ form is also an idealisation. Real institutions confronting visible scarcity typically change their instrument set — buyouts, engineered transfers, new infrastructure — rather than hold a single response law with constant delay. Any estimated τ therefore summarises a changing control architecture, not a structural constant.

Box 1. Management regimes as sample-and-hold architectures.

Four institutional regimes in the review–hold vocabulary.

Regime	Review	Instrument	Hold	Emergency clause
Annual TAC fishery	1 yr	TAC (quota)	1 yr	in-year adjustment where provided
Multiannual plan	3–5 yr	fishing-mortality rule	multi-year	trigger review
Moratorium	indefinite	closure	until review	reopening rule (Section 3.8)
Data-limited system	irregular	effort controls	variable	crisis response

Notation. Throughout, N denotes the resource stock, Z the institutional deficit signal, and E the extraction effort. The review Poincaré map — the discrete-time map obtained by sampling the continuous flow at review instants — is written $\mathcal{P}_{T_r}$; its Jacobian at the fixed point X^*, the monodromy matrix of the sampled loop, is $D\mathcal{P}_{T_r}(X^*)$ (Section 2.3's multiplier condition is written on it); the projection onto the admissible effort interval $[0, E_{\max}]$ is $\Pi_{[0,E_{\max}]}$. The signal map Φ is a nonnegative functional of the deficit, with memory timescale $\tau_m > 0$; its softplus realisation is Φ_k, with shift $\delta = \log 2/10 \approx 0.069$ and sharpness k (the shift is the equilibrium memory level, distinct from and unrelated to the effort-law gain δ_0, as the display below states). The effort law is F_B. The assessment operator is $\mathcal{A}_n$ and the observation operator is $\mathcal{O}$. In the control sections $S(\cdot)$ is surplus production — $S(N)$ on the logistic plant (defined after equation (2)) and $S(A)$ on the stage plant (Section 3.4) — while g (Section 3.3), C_E and C_Z (Section 3.4), and the four-state loop (Section 3.7) are defined where they are used. In the northern cod case (Sections 2.7 and 3.8, Table 2) S is instead spawning stock biomass (reported as SSB in Table 2), $\mathfrak{s}$ the unstable threshold, K the unexploited carrying capacity, $C(t)$ removals, M instantaneous natural mortality (with extra mortality M_x in Lemma 2.2), and F fishing mortality. The two scopes of S never share an equation, and no other symbol serves two sorts: the monodromy is written $D\mathcal{P}_{T_r}(X^*)$ throughout, and Lemma 2.2's production maximum is written $f_{\max}$, leaving M to natural mortality alone. Symbols are tabulated with their definition sites in Table 4 of Appendix A.

Let review times be $t_n = nT_r$. Between reviews, extraction effort is held at E_n and the resource stock follows the logistic process under held effort:

$$\dot{N}(t) = rN(t)\left(1 - \frac{N(t)}{K}\right) - qE_nN(t), \qquad t \in [t_n, t_{n+1}), \tag{1}$$

while the institutional signal Z is the filtered nonnegative deficit

$$\dot{Z}(t) = \frac{\Phi\big(qE_nN(t) - S(N(t))\big) - Z(t)}{\tau_m}, \qquad \tau_m > 0, \tag{2}$$

with $S(N) = rN(1 - N/K)$ the surplus production and $\Phi \geq 0$ a nonnegative signal map with memory timescale τ_m. The assessment available at review n is

$$\widehat{Z}_n = \mathcal{A}_n(\{Y_j : t_j \leq t_n - \tau_{\text{dec}}\}), \qquad Y_j = \mathcal{O}\big(N(t_j)\big) + \varepsilon_j, \tag{3}$$

which separates the latent process N, the observations Y, and the assessments $\widehat{Z}$. Measurement and assessment errors need not be independent or Gaussian, but the admissible error support is restricted to $\widehat{Z}_n \geq 0$. The multiplicative-error robustness experiment of Section 3.3 satisfies this by construction. In the executed calibration all parameters entering the effort law are positive (r, K, q, $E_{\max}$, Δ_{ref}, Z_{ref}, $\tau_m > 0$); because $Z_{\text{ref}} > 0$ and $\widehat{Z}_n \geq 0$, the rational term is well defined on the admissible state space. The effort law's rational term is undefined at $\widehat{Z}_n = -Z_{\text{ref}}$ and changes sign beyond it, and projection of the final command does not repair an undefined update. In the baseline deterministic record the assessment operator is exact and contemporaneous, $\widehat{Z}_n = Z(t_n^-)$: each command uses the just-flowed end-of-period state, indexed by the review that reads it ($\widehat{Z}_{n+1}$ in (4)); the exact update holds this same flowed value. The review map is the projected forward-Euler step

$$E_{n+1}^{\text{cmd}} = \Pi_{[0,E_{\max}]}\left\{E_n + T_rF_B(E_n, \widehat{Z}_{n+1})\right\}, \tag{4}$$

— one specific reviewed controller among many (direct command rules, exact integration under held assessment, incremental rules with fixed step). Because the increment scales with T_r, a sweep in T_r changes both the hold duration and the accumulated gain per review. The separating computation is the exact held-assessment update (the exact update), reported in Section 3.4 with attribution to the companion delay study (Abaee, 2026). The effort law is

$$F_B(E, Z) = \left(1 - \frac{E}{E_{\max}}\right)\left[\eta E\left(\frac{Z}{\Delta_{\text{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0\frac{Z}{Z_{\text{ref}} + Z}\right],$$

The protective comparator is the quota-tracking law

$$F_B^{\text{prot}}(E, Z) = \left(1 - \frac{E}{E_{\max}}\right)\eta_p\big(E_{\text{cap}}(Z) - E\big), \qquad E_{\text{cap}}(Z) = E_0\frac{Z_{\text{ref}}}{Z_{\text{ref}} + Z},$$

with $\eta_p = \eta$ and $E_0 = E^*(Z_{\text{ref}} + \delta)/Z_{\text{ref}}$, calibrated so its unique interior rest coincides with the extractive fixed point; its linearised gains there are $C_E = -0.850336$, $C_Z = -1.661702$ (Abaee, 2026, equation (3)), reviewed through the same projection $\Pi_{[0,E_{\max}]}$. At $E = 0$ the law commands $\eta_p E_{\mathrm{cap}}(Z) > 0$, so zero effort is not absorbing; saturation at $E_{\max}$ is through the gate and the projection.

together with the shifted, floored softplus signal map

$$\Phi_k(s) = \max\left\{0,\ \frac{1}{k}\log\bigl(1+e^{ks}\bigr) - \frac{\log 2}{k} + \delta\right\}.$$

with shift $\delta = \log 2/10 \approx 0.069$, the equilibrium memory level ($Z^* = \Phi(0) = \delta$; distinct from and unrelated to the effort-law gain δ_0) and sharpness k — the value used in the computed records, $k = 10$, is stated where it is used in Section 3.4 and collected in Table 3 of Appendix A. The multiplier records of Section 3.4 presuppose a fixed point interior to the nonsmooth regions of Φ_k and $\Pi_{[0,E_{\max}]}$: $\Phi_k(0) = \delta$ gives the equilibrium deficit $s^* = 0$ uniquely, with softplus slope $sp_k'(0) = 1/2$, and $E^* = 2.09$ lies in the interior $(0, E_{\max})$; the coordinates $(89.55188, \delta, 2.08962)$ are listed in Table 3 of Appendix A.

Equations (1)–(4) with $\Phi = \Phi_k$ constitute the extractive controller studied throughout. It is an explicit controller discretisation, not a generic harvest-control rule and not a model of protective quota reduction. Three comparators are distinctly specified: the protective controller (the quota-tracking law above, the same machinery with the response to decline entering with the opposite sign), the fixed-plan controller (no state-responsive update between scheduled resets), and the hybrid controller (annual-change limits, emergency triggers, or legal overrides). The comparators are specified for the prospective management-strategy evaluation (Section 4.5), not presented as completed experiments. A fixed plan and a state-responsive plan with the same nominal review period are not the same intervention. The model-level run of the comparator evaluation on the logistic hold-map core — the plant of the closed-form operator results — is executed and reported in Section 3.4 at the stated evidential status (deterministic, seed-fixed, on the specified architecture only). The prospective real-system design of Section 4.5 remains unexecuted.

2.2 Registration conventions

The deterministic record fixes a one-step flow-then-update convention, stated in pre-review/post-review terms. On $[t_n, t_{n+1})$ effort is held at E_n. At the review instant t_{n+1} the contemporaneous, exact assessment $\widehat{Z}_{n+1} = Z(t_{n+1}^-)$ of the flowed state is read ($\tau_{\mathrm{dec}} = \tau_{\mathrm{dep}} = 0$), and equation (4) computes the command E_{n+1} held on the following interval. The pre-review state is $X_n^- = (N(t_n^-), Z(t_n^-), E_{n-1})$ and the map used is the pre-review Poincaré map $X_{n+1}^- = \mathcal{P}_{T_r}(X_n^-)$. The reversed ordering (command first, flow second) is a different hybrid system and is not used. No decision or deployment queue is present, and multiplicative assessment error enters only the designated robustness experiment. A positive τ_{dep} requires an explicitly specified hold or interpolation rule.

The computational status of the response-region records is stated with the same care. Classification used long-horizon trajectories, tail amplitudes, multiple histories, and integration-step refinement. It did not use Poincaré-map multipliers, monodromy matrices, or Floquet multipliers. The reported bands are therefore finite-grid, trajectory-classified response regions, and no Neimark–Sacker, flip, or other multiplier-crossing classification is claimed. Until the solver configuration and initial histories are attached (Appendix A), the stage-output values carry provisional status. For the logistic hold-map core the multiplier-based boundary calculation is reported in Section 3.4. For the stage-structured map the boundary record is supplied by the labelled reconstruction of Section 3.4 — a newly specified, pre-registered object whose full specification (flow/update ordering, information pattern, derivative construction, multiplier trajectories, nonlinear trajectories, solver configuration) is given there. The original stage record remains unattached, and its output values keep provisional status. The computational requirements of this section and of Sections 2.4, 2.5, 3.3, and 3.4 — solver configurations, initial histories, stock identifiers, the eligibility table, the null-calibration record, simulation code and seeds, and the reconstruction's plan, code, and output tables — are consolidated in Appendix A, which also fixes the pre-registration convention.

2.3 The review-map operators

Three objects are in play: the continuous-delay equation, the logistic hold map, and the stage-structured review map. The continuous delay equation and the logistic sample-and-hold map are the same feedback loop under two delay operators. The stage-structured map is a different ecological plant (delayed-recruitment stage structure against scalar logistic surplus production) with a different state dimension; it is not an operator substitution on the logistic plant. A valid architecture comparison holds the ecological plant, controller linearisation, equilibrium, and parameter vector fixed while changing only the timing operator. The 3–4 yr windows and the ≈ 6.5 yr exact-update crossing below are therefore reported as a comparison across plants and operators, which does not isolate the operator effect; the one-plant operator contrast in Section 3.4 isolates the operator for the logistic plant. Changing the operator can move or delete a crossing, and that relocation is a property of the review map's spectrum, not a refutation of the loop mechanism. The discipline throughout is that every stability statement names its operator, and statements computed on the hold map, on the stage-structured review map, and on the continuous-delay equation do not transfer to one another. The operative condition is $\det(D\mathcal{P}_{T_r}(X^*) - e^{i\theta}I) = 0$ on the map to which the statement refers, with $D\mathcal{P}_{T_r}(X^*)$ the Jacobian of the review map at its fixed point (the monodromy matrix of the sampled loop).

For a deterministic sampled system, the inter-review flow and the review update combine into a Poincaré map $X_{n+1} = \mathcal{P}_{T_r}(X_n)$, where X_n includes the ecological state, memory, held command, and any explicitly modelled queue. A fixed point is locally asymptotically stable if and only if every eigenvalue of $D\mathcal{P}_{T_r}(X^*)$ lies inside the unit disk. A verified crossing through the unit circle as the fixed parameter T_r changes is a sampled-data parameter bifurcation. It is not rate-induced tipping, which requires a nonautonomous parameter path and a loss of tracking as the ramp speed changes (Ashwin, Wieczorek, Vitolo, and Cox, 2012).

2.4 The cross-sectional spectral screen

The screen's input layer is a fixed 42-stock cohort of the RAM Legacy Stock Assessment Database v4.66 (Ricard, Minto, Jensen, and Baum, 2012): the annual-review designation (applied from general regime knowledge, not per-stock primary verification) covers the full 58-stock panel, and 42 stocks pass the $n \geq 20$ valid-point filter (series lengths 20–77 yr). The release records stock and assessment metadata and series such as biomass, fishing mortality or exploitation rate, total allowable catch, catch advice, and effort. It does not encode controller sign, the decision rule, or dated decision and deployment queues. The annual-review designation is therefore a criterion applied in this analysis, not a database classification of a stock as extractive or protective.

For each eligible biomass (spawning-stock biomass) series, the analysis removes a linear trend, computes a Lomb–Scargle periodogram (Lomb, 1976; Scargle, 1982), normalises it to unit total power, and integrates power in prespecified bands: target bands A (2.5–5 yr, anchovy class) and B (5–9 yr, sprat class), bracketing the observable-specific dominant biomass peaks near 4 and 8 yr of the archived stage-map diagnostics (provisional records, Section 3.3), with contextual bands C (9–14 yr, cohort/recruitment) and D (14–30 yr, trend/regime). No multiplier-crossing verdict depends on the archived diagnostics. The tested statistic is band-integrated power, compared with a per-series AR(1) red-noise null (coefficient by lag-1 autocorrelation of the linearly detrended series; 200 replicates, seed 7, detrending inside each replicate; Lomb-Scargle on 1,500 frequencies from $1/(2 \cdot \text{span})$ to 0.5 yr^{-1}). Empirical p-values over the 84 target-band cells (42 stocks by bands A and B) are adjusted by the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995), which controls the false discovery rate, not the familywise error rate. Dependence across stocks (shared climate forcing, shared assessment methods) is acknowledged (Benjamini and Yekutieli, 2001). The reported zero count is the BH-adjusted result. A peak is classified as robust only if it survives the null comparison and the multiplicity adjustment. The cohort table, screen routine, and multiplicity reconstruction (`ram_target_stocks.csv`, `ram_crosssection.py`, `verify_bh.py`) are deposited with the article. The screen is a restricted result: its weight is limited to the refutation stated in Section 4.2, and its null verdict survives the eleven-variant sensitivity battery of Supplementary S9.1.

2.5 Power analysis

Power experiments inject the model-generated effort signal into white noise (multiplicative log-normal, fractional σ) and apply the same band-power statistic. Formally the reported power is $P(\text{reject} \mid \text{injected signal})$ at the 5% level over the design grid (the conventional power-analysis framing of Cohen, 1988). Power is estimated on 100–200 yr synthetic records across noise scales. The demonstration routine and grid driver with the executed configuration and seeds (`power_demo.py`, `power_driver_v2.py`) are deposited with the article. The executed grid configuration is Supplementary S13.1.

2.6 The structured case search

The case search is a curated inventory, not a systematic review. It considered more than thirty systems spanning fisheries, aquaculture, groundwater, surface water, rangeland, wildlife harvest, forestry, and produced-capital markets, under four criteria:

(i) a responsive institutional feedback rather than a one-time cap or ban;
(ii) an independently dateable response or implementation lag;
(iii) no major environmental, biological, or structural driver empirically supported as generating the same outcome, where the comparative rule is that a candidate institutional mechanism must outperform pre-registered alternatives on held-out prediction, phase ordering, or intervention response (since "capable of generating" is not a falsifiable exclusion in open ecological systems);
(iv) individual-resource or individual-station data rather than an aggregate series.

Behind criterion (iii) is an explicit list of the alternative mechanisms that can mimic or replace an institutional cycle: stage/maturation delays and cohort resonance; recruitment suppression versus stock culling; support-pool/nutrient limitation; pollution/waste feedback; predator–prey or climatic forcing; and direct abiotic mining and slow-store exhaustion. The rule is that a candidate institutional mechanism must be compared with these alternatives rather than identified from periodicity alone. No system in the searched set meets all four criteria; the case-screening table (32 systems) and query log are recorded in Supplementary S4.

2.7 The northern cod case: constitutive model and evidence

The northern cod case (NAFO 2J3KL) is treated as a bounded empirical object. Its illustrative constitutive model is the strong-Allee surplus equation:

$$\frac{dS}{dt} = rS\left(1 - \frac{S}{K}\right)\frac{S - \mathfrak{s}}{K - \mathfrak{s}} - C(t), \tag{5}$$

with S the spawning stock biomass, r the intrinsic growth rate, K the unexploited carrying capacity, $\mathfrak{s}$ the unstable threshold, and $C(t)$ the removals. The Schaefer model is the formal limit of this family ($\mathfrak{s} \to -\infty$) in which the factor $(S - \mathfrak{s})/(K - \mathfrak{s})$ is replaced by 1. No value of $\mathfrak{s}$ makes the displayed factor identically 1 (it tends to 1 only in the limit $\mathfrak{s} \to -\infty$), so the Schaefer comparison is a change of growth law, not a parameter specialisation. The constitutive equation is illustrative, not a fit — and the obstruction class is the one-dimensional autonomous equation with fixed parameters and fixed removals. The displayed equation is autonomous exactly when removals are fixed.

Remark 2.1 (Phase-line obstruction). Let $f : \mathbb{R} \to \mathbb{R}$ be locally Lipschitz, and let $x(\cdot)$ be a nonconstant solution of the scalar autonomous ODE $\dot{x} = f(x)$. Then x is strictly monotone; it never attains an equilibrium value and cannot cross an equilibrium in either direction. Consequently, an exact path that repeatedly rises and falls across a common interval is incompatible with any single fixed scalar autonomous model.

Proof. If x is nonconstant on an interval and $x(t_1) = x(t_2)$ with $t_1 < t_2$, then by Rolle's theorem there exists $\tau \in (t_1, t_2)$ with $\dot{x}(\tau) = 0$, i.e. $f(x(\tau)) = 0$. By uniqueness the solution through $(\tau, x(\tau))$ is the constant equilibrium solution, a contradiction. Hence every solution is strictly monotone or constant. If a trajectory met an equilibrium value x^* at any time, it would be constant thereafter and, by uniqueness applied backwards, before; a nonconstant solution therefore never attains an equilibrium value and cannot cross one in either direction. A path that rises and falls across a common interval attains some level twice with opposite directions, contradicting monotonicity. $\square$

Lemma 2.2 (Extra-loss threshold shift, conditional).

Statement. Assume $0 < \mathfrak{s} < K$. (i) Constant loss. In the constitutive model with an additional constant removal $C > 0$, the production function is replaced by $f_C(S) = f(S) - C$; the positive equilibria of the modified function are the solutions of $f(S) = C$, the smaller one (the effective threshold) lies to the right of $\mathfrak{s}$, and at the production maximum the two positive equilibria coalesce and beyond it disappear — all conditional on the solutions existing, i.e. on C being below the production maximum. For $C = 0$ the smaller positive equilibrium is $\mathfrak{s}$; the strict rightward shift holds for $C > 0$.

(ii) Proportional mortality. With extra mortality $M_x > 0$ the modified production is $f_M(S) = f(S) - M_x S$; positive equilibria solve $f(S)/S = M_x$, $S = 0$ remains an equilibrium (unlike case (i), where $\dot{S}|_0 = -C < 0$ drives the model below zero unless the removal is constrained near zero or the state equation is modified), and the smaller modified positive root lies strictly above $\mathfrak{s}$ (in $(\mathfrak{s}, S_h)$, where S_h is the vertex of the per-capita curve $h(S) = f(S)/S$).

Proof. The constitutive production $f(S) = rS(1 - S/K)(S - \mathfrak{s})/(K - \mathfrak{s})$ vanishes exactly at 0, $\mathfrak{s}$, K; it is negative on $(0, \mathfrak{s}) \cup (K, \infty)$, positive on $(\mathfrak{s}, K)$, and its derivative vanishes exactly at

$$S_\pm = \frac{K + \mathfrak{s} \pm \sqrt{K^2 - \mathfrak{s}K + \mathfrak{s}^2}}{3}, \qquad 0 < S_- < \mathfrak{s} < S_+ < K,$$

so f has exactly one critical point in $(\mathfrak{s}, K)$ — the unique maximum $S^* = S_+$ with value $f_{\max} = f(S^*) > 0$ — and f is strictly increasing on $(\mathfrak{s}, S^*)$ and strictly decreasing on (S^*, K).

(i) The equilibria of $f_C(S) = f(S) - C$ solve $f(S) = C$, so none lie outside $(\mathfrak{s}, K)$, where $f \le 0 < C$. For $0 < C < f_{\max}$, the intermediate value theorem on $[\mathfrak{s}, S^*]$ ($f(\mathfrak{s}) = 0 < C < f_{\max} = f(S^*)$) and on $[S^*, K]$ ($f(K) = 0 < C < f_{\max}$) supplies one root in each interval, and strict monotonicity makes each root unique. The smaller root satisfies $S_1(C) \in (\mathfrak{s}, S^*)$ — the effective threshold lies strictly to the right of $\mathfrak{s}$. As $C \uparrow f_{\max}$ the two roots approach S^* from either side (coalescence), and for $C > f_{\max}$ the equation $f(S) = C$ has no solution: the two positive equilibria have disappeared.

(ii) Write $f_M(S) = f(S) - M_x S = S\,(h(S) - M_x)$ with $h(S) = f(S)/S = \frac{r}{K - \mathfrak{s}}(1 - S/K)(S - \mathfrak{s})$, a quadratic with roots $\mathfrak{s}$ and K, negative on $(0, \mathfrak{s}) \cup (K, \infty)$, and unique maximum $H = h(S_h)$ at the vertex $S_h = (\mathfrak{s} + K)/2$. Then $f_M(0) = 0$: zero remains an equilibrium in every case. The positive equilibria solve $h(S) = M_x$; for $0 < M_x < H$ the same two-interval argument gives exactly two roots, the smaller in $(\mathfrak{s}, S_h)$ — the threshold shift — with coalescence at $M_x = H$ and disappearance for $M_x > H$, where $f_M(S) = S(h(S) - M_x) < 0$ on $(0, \infty)$.

In case (i), by contrast, $f_C(0) = -C < 0$, so $\dot{S}|_0 = -C$ drives the state below zero unless the removal is constrained near zero or the state equation is modified — as claimed. □

The case evidence is the assessment record (the recovery-potential and sequential stock assessments: DFO, 2011, 2016, 2022, 2024). The crash window (1991–1995) and the subsequent windows are documented by the assessment table of DFO CSAS SAR 2016/026 (Table A2), produced by the Northern Cod Assessment Model (NCAM; Cadigan, 2016), together with the dated governance events (the moratorium announcement of 2 July 1992; the reopening of 26 June 2024 with a total allowable catch of 18 kt). The survival column $\exp(-M)$ is a transformation of the reported instantaneous mortality estimate, not an independently observed survival series.

3 Results

3.1 Forward invariance of the sampled process

A first well-posedness check on the sampled controller is that the state space on which the sample-and-hold dynamics act — nonnegative stock, nonnegative memory signal, admissible effort — is preserved by the review map. This check comes first because every later result relies on the state remaining inside this set between reviews.

Proposition 3.1 (Forward invariance of the sampled process). Assume:

(H0) $r, K, q, E_{\max}, \tau_m, Z_{\text{ref}}, \Delta_{\text{ref}} > 0$.

(H1) $N(0) \geq 0$, $Z(0) \geq 0$, $E_0 \in [0, E_{\max}]$.

(H2) The signal map Φ is non-negative.

(H3) Every effort command is projected by $\Pi_{[0,E_{\max}]}$.

(H4) Any between-review deployment interpolation remains in the convex interval joining consecutive commands. The main model takes $E(t) = E_n$, which satisfies (H4) trivially; the proposition is stated for the hold-or-interpolate class.

Then $N(t) \geq 0$, $Z(t) \geq 0$, and $E(t) \in [0, E_{\max}]$ for every time at which the sampled solution exists.

Proof. Proceed by induction over review intervals. Suppose that at a review time t_n one has $N(t_n) \geq 0$, $Z(t_n) \geq 0$, and that the command to be held or interpolated lies in $[0, E_{\max}]$. Projection places the next command in the same closed interval; a hold retains an endpoint of that interval, and any convex interpolation $E(t) = (1 - \theta(t))E_n + \theta(t)E_{n+1}$ with $0 \leq \theta(t) \leq 1$ also remains in $[0, E_{\max}]$. On $[t_n, t_{n+1}]$, Eq. (1) gives

$$N(t) = N(t_n) \exp\left\{ \int_{t_n}^{t} \left[r \left(1 - \frac{N(s)}{K} \right) - qE(s) \right] ds \right\} > 0$$

for a positive initial value, while if $N(t_n) = 0$, uniqueness gives the identically zero solution on the interval; N cannot become negative. Set $\nu(t) = \Phi(qE(t)N(t) - S(N(t))) \geq 0$. Variation of constants in Eq. (2) with $\tau_m > 0$ gives

$$Z(t) = e^{-(t-t_n)/\tau_m} Z(t_n) + \frac{1}{\tau_m} \int_{t_n}^{t} e^{-(t-s)/\tau_m} \nu(s)\, ds \ \geq 0.$$

Boundedness from above holds automatically on the stock equation and is used in the induction: on $[t_n, t_{n+1})$ the stock solves the logistic equation under held effort, and $\dot{N} \leq 0$ at $N = K$ (the harvest term is non-negative), so $N(t) \leq \max\{N(t_n), K\}$ throughout the interval. All three inequalities therefore hold through t_{n+1}; they hold at $t_0 = 0$ by hypothesis, and induction proves them on every review interval for which the sampled solution exists. □

Positivity, boundedness, persistence above a threshold, and viability are distinct properties; Proposition 3.1 establishes positivity and effort admissibility only. The institutional implication is modest but essential: the projected Euler step does not push the state outside its specified admissible set on a single review interval, and any later stability or instability claim is therefore a statement about dynamics inside that set, not about escape from it.

3.2 The rapid-review limit and what it does not establish

Define $u = (N, Z, E)$ and the continuous no-delay system $\dot{u} = G(u)$ assembled from Eqs. (1)–(2) with the effort law evaluated continuously. Suppose G is continuously differentiable with bounded derivative on a compact neighbourhood containing the compared trajectories on $[0, T]$, assessments equal the contemporaneous state, no additional decision or deployment queue is present, projection is inactive, and the sampled and continuous systems share the initial state. The frozen-effort flow followed by the explicit effort step then has a one-step defect of order $O(T_r^2)$ relative to the exact flow of G. The usual discrete Gronwall estimate yields an $O(T_r)$ review-time error and uniform convergence on every fixed finite horizon as $T_r \to 0$ (the sampled-data approximation framework of Nešić and Teel, 2004).

This is a numerical-approximation property, not a finite-review stability theorem. It gives no uniform-in-time estimate, does not preserve stability automatically, and does not control any trajectory-classified response region reported below. It excludes active projection, delayed or erroneous assessment, a deployment queue, and stochastic updates. In particular, it neither identifies a finite T_r with a continuous discrete delay nor implies that sufficiently small positive T_r is stable when the continuous target is unstable. The converse caution is equally important. Under the additional assumptions of hyperbolicity of the continuous equilibrium and C^1-consistency of the sampled map with projection inactive locally, the multipliers satisfy $\mu_j(T_r) = 1 + T_r\lambda_j(A) + O(T_r^2)$ (Stuart and Humphries, 1996), so local stability persists for all sufficiently small positive review intervals provided every continuous eigenvalue satisfies $\Re\lambda_j < 0$; if any $\Re\lambda_j > 0$, the sampled equilibrium inherits instability for sufficiently small T_r. The finite-horizon result therefore neither proves nor refutes stability transfer; the transfer hypotheses are what any such claim would have to supply. The institutional reading is that "rapid review" is not a stability guarantee — a governance loop that reviews frequently approximates a continuous controller only on finite horizons, and the question of whether short review intervals stabilise or destabilise the equilibrium is decided by the spectrum of the map, not by the rapid-review limit.

3.3 Response regions of the delayed-recruitment review maps

The delayed-recruitment records, classified from long-horizon trajectories on the stage-structured review map, locate exploratory response regions by stock class (the delayed-recruitment oscillation literature of Gurney, Blythe, and Nisbet, 1980, supplies the classical reference point). These are archived records whose generating computation was never attached (Section 2.2): they are reported below at provisional status, and the pre-registered stage-map reconstruction of Section 3.4 does **not** reproduce the multi-year windows on the anchovy- or sprat-class parameterisation of that plant (reporting convergence there, with a separate long-horizon band that has no archived counterpart). The regions below are therefore archived, unreproduced records, not results of the reconstructed object:

- **Anchovy-class:** persistent tail oscillation near $T_r \approx 3$–4 yr, with a weak response at $T_r = 2$ yr; annual-review trajectories converge over the tested grids for every tested effort-response value.
- **Sprat-class:** persistent tail oscillation near $T_r \approx 6$–12 yr.
- **Cod-class:** trajectories converge to equilibrium for every tested $T_r \in [1, 20]$ yr, although the corresponding continuous-delay calculation has an oscillatory interval — a convergence-over-tested-grid record, not a stability theorem for every history.

- **Slow-stock class:** for $r \in (0.01, 0.05)$ yr^{-1}, oscillation over part of the tested grid below approximately 20–30 yr and convergence at longer review intervals, with transition brackets between approximately 30 and 50 yr depending on r. This one-sided pattern does not contradict rapid-review consistency: the continuous no-delay target is itself unstable over part of the slow-r regime, so small-T_r trajectories may approximate an unstable target on finite horizons. The record is conditional on the delayed-recruitment variant and is not a general claim that slower review stabilises governance; the dominant timescales are centuries (the archived slow-stock cohort cycle: $P \approx 250-360$ yr), beyond the length needed to resolve multiple cycles in most institutional records.

The corresponding continuous-delay calculations on the delayed-recruitment parameterisation — g is its maturation delay, and the response regions are located through the product rg — locate response regions near $rg \approx 1.5$–1.6: for $g = 2$ yr, $r \in (0.77, 0.81)$ yr^{-1} at $\eta = 0.914$ (the uncalibrated Candidate-A baseline; Appendix A) with a delay interval of approximately 2.6–7.8 yr; for $g = 1$ yr, the high-r interval is approximately 1.565–1.585 yr^{-1} with a delay interval of 1.6–3.5 yr. These are finite-grid trajectory summaries, not closed analytical stability regions.

A τ_0 decomposition separates, within the delayed-recruitment operator, cohort-driven from institutional-delay-induced crossings: at $g = 5$ yr and $\eta = 0.914$ the raw r-window [0.0076, 0.3705] narrows to the institutional-only [0.2660, 0.3285], and fish-range crossings from 12 to 8, under τ_0-stable classification by nonlinear ground truth. The separation is within one operator; it does not isolate the cross-operator effect compared in Section 4.1. The $g = 5$ middle band overlaps slow cod-class growth rates but remains confounded with cohort resonance. That operator differs from Section 3.7's four-state loop. Full windows, the twelve-cell decomposition grid, and the middle-band table are tabulated in Supplementary S9.2.

Multiplicative assessment-error experiments preserve the anchovy-class trajectory region through 30% error and produce no noise-induced persistent tail oscillation at annual review in the tested ensemble. This is a robustness summary for the specified multiplicative perturbation only. The archived diagnostics carry observable-specific dominant peaks — approximately 4 yr in anchovy-class biomass with 12 yr in effort, and approximately 8 yr in sprat-class biomass with 60 yr in effort. These are reported only as observable-specific dominant peaks over the analysed windows: components of one stationary periodic orbit cannot have different fundamental periods, and harmonics, subharmonics, modulation, or transient spectral content are unresolved; no decomposition has established whether the baseline term, signal regularisation, or another controller component dominates these responses. The archived amplitudes indicate percent-scale biomass excursions and order-one effort excursions relative to equilibrium. The exact percentages are not used as effect-size estimates, because the archived record does not distinguish peak-to-peak range from half-range and the approach to the large effort response contains a long transient.

3.4 The operator spectra

The logistic hold-map core permits a closed-form expression for the held logistic flow; the linearised review map is then constructed from this flow, the signal dynamics, and the update derivatives. With effort held at E, the logistic flow over one review interval is, for $a(E) = r - qE \neq 0$,

$$N(t_{n+1}^-) = \frac{a(E)\, N(t_n^-)\, e^{a(E)T_r}}{a(E) + \frac{r}{K} N(t_n^-)\big(e^{a(E)T_r} - 1\big)},$$

and $N(t_n^-)\big/\big(1 + \frac{r}{K} N(t_n^-) T_r\big)$ for $a(E) = 0$. In computations, the limit branch is used when $|a(E)| < 10^{-12}$. The signal flow enters through equation (2) with this forcing, and the linearised pre-review monodromy is assembled from the held-flow state transition, the assessment derivative at $Z(t_{n+1}^{-})$, and the update derivative of equation (4). The exact held-assessment controller — the separating comparator that isolates the Euler step — is the exponential update (the exact update; its review map the exact map) $e_{n+1} = e^{C_E T_r} e_n + \frac{e^{C_E T_r}-1}{C_E} C_Z z_n$ for $C_E \neq 0$, and $e_{n+1} = e_n + T_r C_Z z_n$ for $C_E = 0$, with $e_n = E_n - E^*$ and $z_n = Z_n - Z^*$ the deviations from the compared fixed point — the exact solution, over one review interval, of the linearised effort law $\dot{e} = C_E e + C_Z z$ with the flowed end-of-period assessment z held, where C_E and C_Z are the coefficients of that linearisation (the partial derivatives of F_B with respect to E and Z at the compared fixed point; unrelated to the removals C of the cod case), and equation (4)'s increment is the forward-Euler discretisation of the same linear object. This comparator is analytical, not a proposed institutional rule. Comparing its monodromy with the forward-Euler map's separates review cadence, zero-order holding, and Euler discretisation. That comparison is reported here: executed and verified on the companion delay study's identical hold map (Abaee, 2026; same effort-law bracket, same softplus signal, same forward-Euler monodromy reproducing the 47.536 yr crossing reported here), the exact-update annual spectral radius is $\rho = 1.00035$ (Euler 1.00055); the 47.536 yr and 79.143 yr crossings are command-step artefacts of the forward-Euler update; the exact map's single unit-circle crossing is ≈ 6.5 yr; and the restabilising direction is preserved.

On the logistic hold-map core, the undelayed equilibrium is already unstable, so annual review is unstable. On the computed crossing set, the sampled equilibrium has a complex unit-circle crossing — the spectral signature of a Neimark–Sacker bifurcation, nonlinear nondegeneracy not verified — at $T_r^{\mathrm{UC}} = 47.536$ yr under the forward-Euler command step and at ≈ 6.5 yr under the exact update, with the same restabilising direction. For an institution implementing the incremental Euler rule these crossings are dynamically real; they are artefacts only relative to the exact update.

The complete crossing record. On the logistic hold-map core the unit-circle crossing record over $[0.2, 200]$ yr is complete (Figure 2). It is computed from the linearised pre-review monodromy on a 200,001-point scan with bisection refinement (deterministic; it reproduces the numbers reported here on the range they cover). The forward-Euler update crosses twice on the extractive channel — a complex pair at $T_r = 47.536$ yr (unstable $\to$ stable) and a real -1 multiplier at 79.143 yr (stable $\to$ unstable), stable band $[47.536, 79.143]$ yr — and once on the protective channel — a real -1 at 2.306 yr (stable $\to$ unstable), stable only on $[0.2, 2.306]$ yr. The exact update crosses once on the extractive channel — a complex pair at 6.501 yr (unstable $\to$ stable) at the baseline parameter vector (Table 3), stable on $[6.501, 200]$ yr — and never on the protective channel, which is stable at every tested interval with maximum spectral radius 0.9967. Both Euler crossings of the extractive channel are therefore command-step artefacts relative to the exact map's single crossing, and the Euler protective instability beyond 2.306 yr is likewise an artefact: the exact protective map is stable throughout.

Spectral margins of the annual-review verdict. The annual-review instability is a near-unit-circle record, and its margins are part of the record. The exact update's annual spectral radius is $\rho = 1.00035$ — a modulus exceeding unity by 3.5×10^{-4} — with the forward-Euler update at 1.00055 and the protective channel at 0.9838 (maximum 0.9967 over the tested range). The multiplier types are reported with the crossings — a complex pair at the exact map's single 6.501 yr crossing (the Neimark–Sacker signature) and at the Euler extractive crossing (47.536 yr), and real -1 multipliers at the Euler crossings (79.143 yr extractive, 2.306 yr protective) — while the dominant multiplier's angle θ at the crossing and the continuous eigenvalue λ to which Section 3.2's transfer relation refers are listed in Table 3 (Appendix A). At a margin of this size the verdict is conditional on the monodromy's numerical construction — the stated precision record is the 200,001-point scan with bisection refinement; the linearisation construction and floating-point detail are archive items — and on the parameter vector listed in Table 3. What the margin does not affect is the ordering across updates and channels (exact 1.00035 below Euler 1.00055; protective 0.9838 below both) or the crossing directions;

the annual-instability verdict is read at that status. Near-unit-circle behaviour of this size is read as high sensitivity of the verdict to the parameter vector, not as a robust claim about annual-review institutions in general.

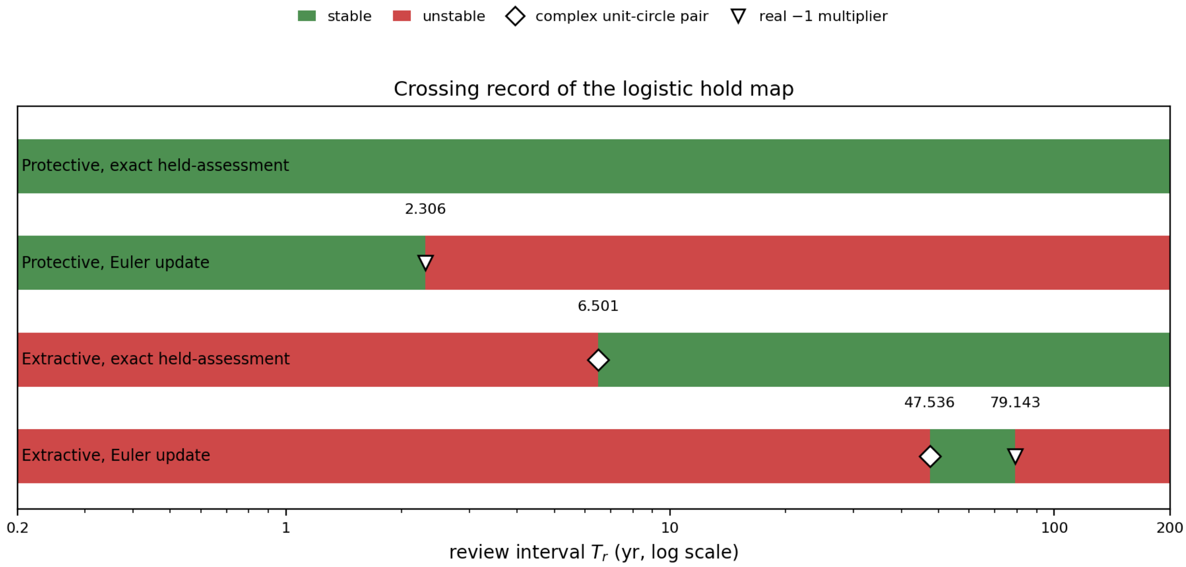

Figure 2: Crossing record of the logistic hold map: stable (green) and unstable (red) review-interval ranges for the four update × channel combinations (forward-Euler and exact updates on the extractive and protective channels), with crossing markers (diamond: complex unit-circle pair; triangle: real -1 multiplier). Annual review is nominally unstable on both extractive updates ($\rho = 1.00035$ exact, 1.00055 Euler; protective 0.9838), conditional on the numerical construction and parameter vector (Spectral margins paragraph); the Euler-reported 47.536 and 79.143 yr crossings are command-step artefacts relative to the exact update's single 6.501 yr crossing, and the protective Euler crossing at 2.306 yr is likewise an artefact — the exact protective map is stable at every tested interval (maximum spectral radius 0.9967).

The one-plant operator contrast. With the crossing record complete, the operator comparison can be made on one plant rather than across plants. The same extractive loop under continuous delay carries the companion delay study's certified Hopf pair at 3.666 and 150.358 yr (Abaee, 2026); the sampled operator's exact crossing sits at 6.501 yr. The companion's window is delay-stabilising, not delay-destabilising: the undelayed loop is already unstable, the lower crossing at $\tau_- = 3.666$ yr is stabilising, the upper crossing at $\tau_+ = 150.358$ yr destabilising, and the loop is linearly unstable for $0 < \tau < \tau_-$ (Abaee, 2026). Annual review is nominally unstable under sampling ($\rho = 1.00035$, conditional on the numerical construction and parameter vector as stated above), and the same loop under continuous delay is likewise unstable at $\tau = 1$ yr; at $T_r = 8$ yr both operators are stable (the sampled loop restabilised above its 6.501-yr crossing, the continuous loop inside its stabilised window). The operator contrast on one plant is an edge contrast: the sampled operator's restabilising crossing (6.501 yr) sits above the continuous operator's stabilising entry (3.666 yr), and the sampled loop's stability persists to the 200-yr scan ceiling where the continuous loop has already destabilised at τ_+ — the operator moves both edges of the stability window on the same plant, and the cross-plant observation of Section 2.3 is a same-plant one.

The undelayed limit, reconciled explicitly. Three statements of this manuscript bear on the undelayed loop and are read together. Section 2.3 states the continuous-delay equation and the logistic hold map to be the same feedback loop under two delay operators; Section 3.2 states that under hyperbolicity the sampled multipliers satisfy $\mu_j(T_r) = 1 + T_r\lambda_j + O(T_r^2)$, so that small-$T_r$ sampled stability follows the sign of $\operatorname{Re}\lambda_j$; and this section records that the undelayed equilibrium of the hold-map core is already unstable (annual $\rho = 1.00035$). The companion's undelayed record fixes the remaining sign: the undelayed linearisation of the same loop violates the Routh–Hurwitz condition, so $\mathrm{Re}\,\lambda > 0$ — the direction the sampled record suggests through Section 3.2's relation, with the sampled instability persisting down to $T_r = 0.2$ yr. Under that sign the records close at the zero-delay limit: the loop is unstable under both operators at zero delay and at $\tau = 1$ yr, and the stabilising crossing at 3.666 yr is the first crossing (there are no crossings in $(0, \tau_-)$; this even count is consistent with the stability-switch principle because the stability state does not change between zero delay and the first crossing) — no additional crossings are required. On this plant, the two operators differ only in where they place the window's edges, as the one-plant contrast above shows. The eigenvalue λ is listed in Table 3; it is a derived eigenvalue, not a parameter, and its sign is read from the companion's undelayed record, and the operator-scoped records stand as stated — annual instability under sampling, instability under continuous delay at $\tau = 1$ yr, and the same-plant edge contrast — with Section 2.3's same-loop statement and Section 3.2's transfer statement consistent with both records.

The protective controller on the same maps. The protective comparator run (sign-reversed response; linearised gains $C_E = -0.850336$, $C_Z = -1.661702$) on the same plant gives exact-update spectral radius 0.9838 at annual review and stability at every $T_r \in [0.2, 200]$ yr (maximum 0.9967). The Euler-run protective instability is an artefact band, not a property of the protective architecture.

Executed comparator evaluation at model level. The protective and fixed-plan comparators are run through the closed loop on the logistic hold-map core (nonlinear plant, $N_0 = 0.95\,N^*$, 800 reviews, metrics on the last 60%, multiplicative assessment error $\{0, 0.3\}$, seeds fixed; the spectral record remains the authoritative result, and the prospective real-system design of Section 4.5 remains unexecuted). Under 30% assessment error with ten seeds per cell: the extractive Euler core crashes to extinction in all ten seeds at $T_r = 8$, 12, and 20 yr, and at annual review its mean depletion frequency (fraction of post-transient reviews with $N < N^*/2$) is 0.22 with mean minimum stock $0.22\,N^*$; the extractive exact update never crashes, with mean depletion falling from 0.07 ($T_r = 1$) to 0.001 ($T_r = 20$); the protective exact update holds the equilibrium at every interval (depletion 0, mean minimum $0.89\,N^*$, rms stock deviation $\leq 0.001K$); the protective Euler update crashes in all ten seeds at $T_r = 20$ yr and shows depletion 0.9996 at $T_r = 12$ — the artefact operating at the trajectory level; in the deterministic baseline with fixed effort set at the equilibrium value, the fixed plan remains at equilibrium — a baseline rest point, not a robust performance result. The linear trajectory distortion between the Euler and exact updates grows with T_r (scaled-norm RMSD ≈ 2 at $T_r = 0.5$ yr, 10^4–10^5 at $T_r = 5$–40 yr, with the Euler path diverging in its artefact bands): the command step is not a small distortion. The stage-map inputs to this evaluation are the archived, unreproduced records of Section 3.3 — annual stability, the anchovy-class 3–4 yr and sprat-class 6–12 yr windows, and robustness to 30% multiplicative assessment error — read at that section's status, not as results of the reconstructed object.

Two consequences follow. First, in the reported simulations relative effort excursions were larger than relative biomass excursions, and any quota-utilisation reading of the signal would require a quota, realised catch, and compliance model, which are not part of the model core. Second, no qualifying example was found in the case search of Section 2.6 of a real small-pelagic system operating a responsive 3–4 yr review, so the window prediction is untested, not falsified. The corresponding qualitative test for slow-regenerating resources is whether a responsive multi-year plan, adjusted against measured change, produces large-amplitude extraction cycles that a frozen multi-year cap does not (Section 4.5).

The stage-map reconstruction record. The stage-output values carry a stated limitation: the stage map's computational record — solver configuration, initial histories, stage registration — is not attached, so those values are not reproducible numerical propositions (Section 2.2). To supply the stage operator's boundary record, a new stage map was specified and pre-registered (plan dated and fixed 2026-09-01, before any run; parameter values and comparison criteria fixed in the plan), and its complete multiplier and trajectory records are reported here. The archived stage-output records are trajectory-classified only (Section 2.2). The reconstruction is a newly specified object: it is not claimed to be the object that produced the archived windows, and the comparison below is a post-hoc consistency check, not a validation of those windows.

Plant. A two-stage delayed-recruitment map with annual steps, $A_{t+1} = s_A A_t + s_J J_t - qEA_t$, $J_{t+1} = f(A_{t-\tau})$ with Beverton–Holt recruitment $f(A) = \alpha A/(1+\beta A)$, $s_A = s_J = e^{-M}$, and the pair (α, β) fixed by steepness and scale: $\beta = (c-1)/A_0$, $\alpha = c(1-s_A)/s_J$, $c = 4h/(1-h)$, $A_0 = 100$ — a scale convention matching the logistic core's carrying capacity. Surplus production $S(A) = s_J f(A) - (1-s_A)A$ replaces the logistic surplus in the deficit signal of equation (2). Every parameter is a cited literature value or a stated convention; none was chosen with reference to the archived windows. The class parameter sets: anchovy $M = 0.90$ yr^{-1} and recruitment delay $\tau = 1$ yr (Pauly and Tsukayama, 1987); sprat $M = 0.40$ yr^{-1} (mid-range of the Baltic SMS keyrun values, 0.3–0.8 yr^{-1}; ICES, 2020) and $\tau = 2$ yr (age at 50% maturity in the Baltic sprat assessment); cod $M = 0.20$ yr^{-1} (the fixed cod-assessment convention; ICES, 2021) and $\tau = 5$ yr (northern cod age at 50% maturity; DFO, 2011); slow-stock $M = 0.045$ yr^{-1} and $\tau = 25$ yr (orange roughy; Branch, 2001). Steepness $h = 0.75$ throughout — the standard default convention when no stock-specific estimate is available — with a sensitivity layer $h \in \{0.6, 0.9\}$. The controller of Section 2.1 is unchanged: equations (3)–(4) in discrete form with T_r varied over $\{1, \ldots, 50\}$ yr at annual internal steps, softplus sharpness $k = 10$ (the comparator machinery's value), contemporaneous exact assessment, and catchability $q = 0.001$ (the hold-map core's value) with a sensitivity case $q = 0.1$ (the archived stage record states no q; both values are reported for the reconstruction). The derivative construction is central finite differences of the review map at the fixed point (step 10^{-6}, chain-rule cross-check to 10^{-4}), and the scan grid is $T_r \in \{1, \ldots, 50\}$ yr at annual internal steps — finite-grid by construction, as the archived record itself was. The extractive channel is primary; the protective channel is computed with quota-tracking gains of Section 2.1 in the linearised review map, the same convention as the logistic protective record.

Records. At the specified controller value every class is stable at annual review — spectral radii $\rho(1) = 0.895$ (anchovy), 0.923 (sprat), 0.956 (cod), 0.994 (slow-stock) — reproducing the archived stage map's annual stability at the tested response value. The protective channel is stable everywhere on the grid (maximum spectral radius 0.968), mirroring the logistic protective record. The extractive channel's only instability at the specified value is a long-horizon band from $T_r = 34$–35 yr to the grid's end for the three faster classes (maximum spectral radius ≈ 1.98 at $T_r = 50$), with no crossings anywhere for the slow-stock class ($\rho(50) = 0.67$); the band is insensitive to the steepness layer. The trajectory classification (2000 review-steps from the specified initial condition — plant at the unfished equilibrium, memory filled, effort at half the equilibrium — tail of 500, relative tail standard-deviation thresholds 2% and 0.1%) agrees with the spectral record for the three faster classes, which converge at every $T_r \in [1, 20]$. In the multiplier-unstable long-horizon band ($T_r \geq 34$ yr) the faster-class trajectories remain biomass-converged or weak (relative tail standard deviation at most 0.6%) while effort excursions grow from approximately 7–15% at band entry to approximately 190% at $T_r = 50$: the instability manifests in the effort channel, to which the classifier — a biomass-tail threshold — is insensitive. The slow-stock class does not agree with its own multiplier record: it shows a persistent oscillation at $T_r = 10$ yr only (relative tail standard deviation 5.5%; dominant period ≈ 24 yr in both stock and effort; effort excursion 537% against a biomass excursion of 18%), converging at every other grid point, against a multiplier record with no crossings anywhere on the grid. A persistent oscillation at a linearly stable fixed point is bistability, a non-decayed transient, or a classification-threshold artefact; the record does not distinguish among them, and the cell is reported as a disagreement between the reconstruction's two records, not as an agreement, and is not used inferentially. Under 30% multiplicative assessment error the anchovy and sprat cells remain converged (maximum relative tail standard deviation 0.0009); the cod cells weaken at $T_r = 10$ and 20 yr (converged $\rightarrow$ weak, maximum 0.0025) while its annual-review cell stays converged (0.0004); and the slow-stock class oscillates persistently at the long-horizon cells $T_r = 30$–50 yr (2.1–2.6%, against a stable multiplier record without error — cells read as noise-driven variance rather than oscillation, since the 2% relative tail threshold is not noise-adjusted) while its $T_r = 10$ cell weakens from its error-free oscillation and its $T_r = 20$ cell weakens (0.017). The catchability sensitivity analysis is informative: at $q = 0.1$ every class is unstable at annual review ($\rho(1) \geq 1.29$); the three faster classes destabilise on $[1, 18{-}31]$ yr, restabilise, and re-enter an unstable band at 36–42 yr, while the slow-stock class is unstable across the entire grid ($\rho(50) = 7.8$) — the reconstructed stage loop's stability profile depends strongly on the catchability scale, which is unspecified for this plant.

Comparison with the archived windows (criteria pre-registered; verdicts from the first complete run):

Criterion (archive)	Reconstruction	Verdict
anchovy persistent oscillation at $T_r = 3$–4 yr	converged at both	MISMATCH
anchovy weak response at $T_r = 2$ yr	converged	MISMATCH
anchovy annual convergence	$\rho(1) = 0.895$	MATCH
sprat persistent oscillation at $T_r = 6$–12 yr	converged throughout	MISMATCH
cod convergence for every $T_r \in [1, 20]$	converged throughout	MATCH
slow-stock oscillation below 20–30 yr, convergence at 30–50 yr	oscillation at $T_r = 10$, convergence at 30–50	MATCH

Reading. The reconstruction reproduces the archived record's qualitative features — annual stability on the stage map, the protective channel's stability, cod-class convergence over the 1–20 yr grid, the slow-stock oscillation-then-convergence pattern, and the effort-versus-biomass excursion ordering — but not the anchovy 3–4 yr or the sprat 6–12 yr response regions: on this specified plant family those classes converge at every review interval, and the only instability the reconstructed loop produces is its own long-horizon band, entering at 34–35 yr and extending to the grid's end, which has no counterpart among the archived windows. The comparison, however, is uninformative rather than a non-reproduction. The stage map states no catchability; the table is computed at the imported value $q = 0.001$; and at the sensitivity case $q = 0.1$ every verdict in the table flips — every class unstable at annual review ($\rho(1) \geq 1.29$), the slow-stock class unstable across the entire grid — so a match or a mismatch at either value is a statement about this specified family at an unspecified scale, not an adjudication of the archived values, whose generating computation is not available. The reconstruction therefore supplies the stage operator's boundary record as a new, fully specified object, whose comparison with the archived windows is a consistency check rather than a validation.

Three verification records qualify the crossing record. First, a one-at-a-time sensitivity battery perturbs each baseline parameter (r, K, q, $E_{\max}$, η, δ_0, Z_{ref}, Δ_{ref}, τ_m, δ) by $\pm 5\%$ and $\pm 10\%$, reproducing all eight committed values before extension (Supplementary S11). The exact-map crossing is highly sensitive to r, K, q, $E_{\max}$, η, and Δ_{ref} — several $\pm 10\%$ cells remove the crossing or move it past 13 yr — moderately sensitive to τ_m, affected by δ (removed at -10%), and insensitive to δ_0 and Z_{ref} (crossing moves < 0.07 yr). The protective channel is stable in every battery cell (maximum $\rho = 0.9971$), so the channel contrast survives uniform $\pm 10\%$ perturbation. The spectral-radius curves for all four update–channel combinations are

Figure 3.

Second, the equilibrium fishing mortality $F^* = qE^*$ organises the q-sensitivity: q and $E_{\max}$ perturbations act on the crossing almost entirely through F^* ($q \times 1.05$ and $E_{\max} \times 1.05$ give $F^* = 0.00219$ with crossings at 13.797 and 13.790 yr respectively; $\times 1.10$ gives 18.377 and 18.367 yr), and a decade-spanning q grid shows review-instability turning on as F^* rises through the baseline 0.00209, with the equilibrium infeasible ($N^* < 0$) once $F^* > r$ (Supplementary S11). Assessed fishing mortality across the 42-stock cohort (median mean-F 0.37) and a 454-stock broader pool (median 0.21) exceeds the illustrative baseline F^* by two orders of magnitude, sitting beyond the model feasibility boundary; quantitative thresholds are therefore scale-conditional (Supplementary S12).

Third, the 6.501 yr crossing is a verified Neimark–Sacker bifurcation of the nonlinear sampled map: the finite-difference Jacobian matches the monodromy to 10^{-6}, the crossing is transversal ($d|\mu|/dT_r = -0.000683$) and non-resonant through fourth order, and the first Lyapunov coefficient is sign-definite across a sevenfold finite-difference step range ($a(0) \approx -0.0008$, supercritical). Long-horizon iterates rotate at the predicted frequency (rotation number 0.027–0.029 against $\theta_0/2\pi = 0.0279$); the invariant circle itself is not directly exhibited because convergence at $|\rho - 1| \sim 10^{-4}$ is too slow for the tested horizons (Supplementary S11).

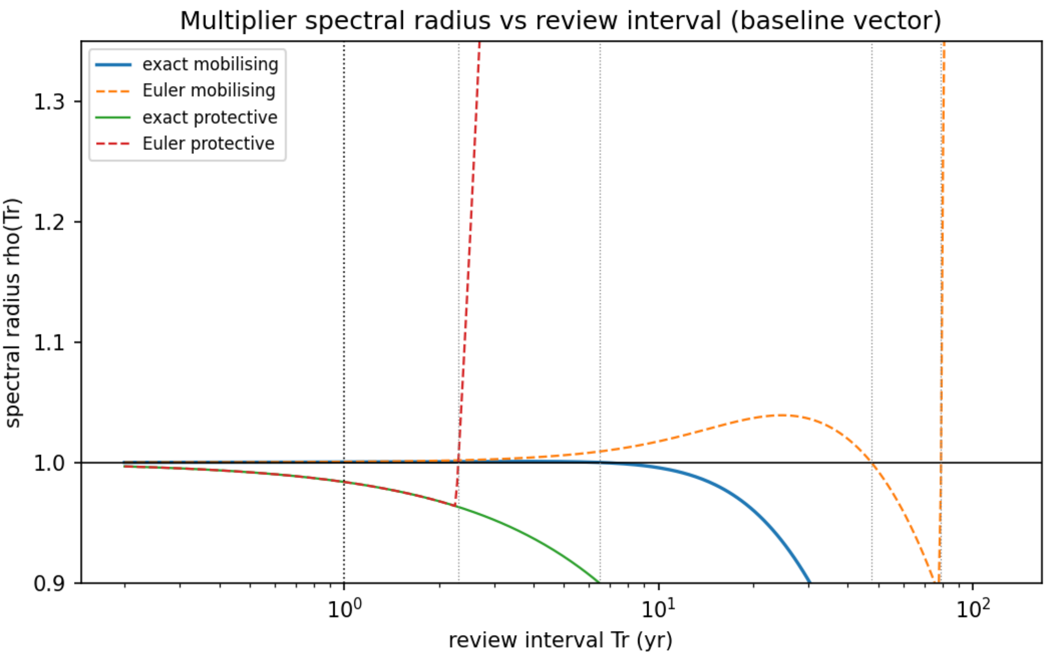

Figure 3: Spectral radius $\rho(T_r)$ against review interval at the baseline parameter vector (Table 3) for the four update–channel combinations. Horizontal line: the unit circle; dotted verticals: the 6.501, 47.536, 79.143, and 2.306 yr crossings and the annual review (1 yr). The exact mobilising curve crosses once; the Euler mobilising curve reports two artefact crossings; the exact protective curve never reaches unity.

3.5 The selected 42-stock spectral screen

No stock in the screened cohort has a peak in the specified biomass target bands meeting all robustness criteria. Baltic sprat, for example, has a biomass coefficient of variation of 0.40, but its variation is low-frequency and regime-dominated rather than a robust target-band peak. The null carries a three-way restriction: it is not proof of absence, not a comparison of controller signs, and not causal evidence that annual review stabilises anything. On the stage-structured review map (Section 3.3), annual-review stability at every tested response value is consistent with this null; consistency is not a test. The institutional reading is therefore limited: under this screen, target-band cycles are not common in assessed-stock records, and any claim that periodicity itself diagnoses an institutional feedback is unsupported by this cohort. That null verdict is robust: an eleven-variant sensitivity battery — the AR(1) baseline plus ARMA(1,1), trend-stationary-no-detrend, circular block bootstrap at three block lengths, Hodrick–Prescott-cycle and first-difference pretreatments, and piecewise-AR(1) regime surrogates at three break dates, each at the published 200-replicate, seed-7 resolution — returns the BH-adjusted zero count in every variant (smallest nominal p 0.0100, Hodrick–Prescott variant; details and the resolution bound in Supplementary S9.1; the screen's restricted-result status is stated in Section 2.4).

The screen's zero count holds under exploratory alternative bands: a broad 2–10 yr band, management-cadence bands (1–3, 3–8, 8–20 yr), and an 8–80 yr effort-proxy band all return zero Benjamini–Hochberg-significant cells, with the effort-proxy band unresolvable (record span below three upper periods) on all 42 records (Supplementary S11). Figure 4 shows the baseline q-value distribution and per-stock null excess. Exploitation intensity carries no spectral excess once the ENSO confound is removed (Spearman 0.03 with Peru/Chile excluded, against 0.31 before; Supplementary S12). Across a 454-stock broader pool, the archived cohort median buffer-years (1.79) sits at the second percentile of resampled medians, independently re-executed (Supplementary S12).

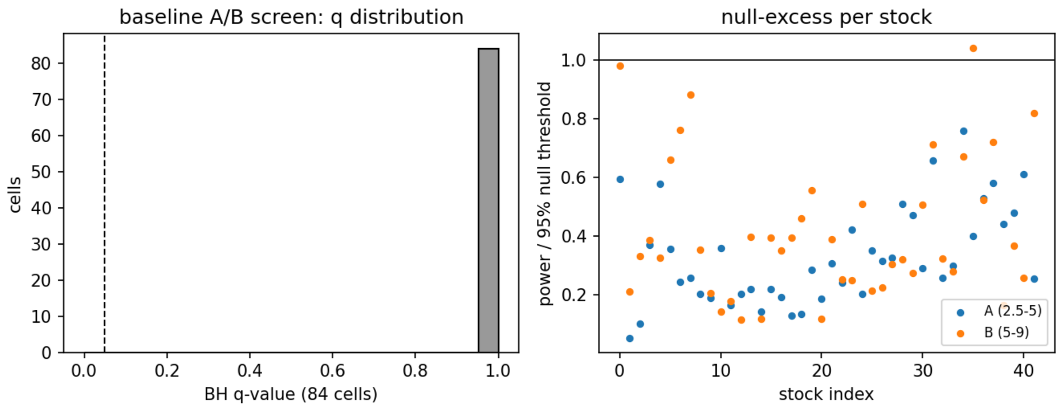

Figure 4: Baseline screen diagnostics. Left: Benjamini–Hochberg q-values over the 84 target-band cells (dashed line: 0.05; no cell significant). Right: per-stock band power relative to the 95% per-stock AR(1) threshold in bands A and B (horizontal line: unity).

3.6 Power of the injected-signal screen

On 100–200 yr synthetic records, the sprat-class signal has estimated power 1.0 at noise scale $\sigma = 0.1$ and approximately 0.24–0.58 at $\sigma = 0.3$. The anchovy-class effort signal has power approximately 0.02–0.14 over the tested horizons and noise levels; its longer effort peak and slow amplitude growth can offset its larger relative excursion. These are conditional simulations: no minimum-power guarantee holds per empirical stock, and the 100–200 yr horizons exceed many eligible series. The anchovy-class null is consequently weakly informative under this test, and the sprat-class result is informative only in favourable noise and record-length regimes. The empirical screen is therefore a selected-cohort consistency check and the baseline for a prospective design, not a general test of the extractive mechanism. The sprat-class 1.0 at $\sigma = 0.1$ is trend detection (growth-transient run-up in the 30–120 yr band), not cycle detection, and is fragile at $\sigma = 0.3$.

The evidentiary separation is complete: the stage-dependent regions, noise experiments, and power estimates do not establish population-wide frequencies or policy effects and cannot be transferred to the extractive hold-map controller or to protective control. The complementary prospective design is specified in Section 4.5 (detail in Supplementary S13.3).

3.7 Structured case search: no qualifying cases

No candidate satisfied all four criteria after primary-source and station-level review. The zero count is an evidential gap, not a disconfirmation: it records that no unconfounded oscillator was found, not that none exists. Bangkok (durably responsive institutional feedback: the groundwater-extraction control regime persisted across multiple review cycles rather than being a one-time cap or ban) and La Mancha Oriental (on the stabilising side before its 2019–2023 extraction relapse) are the closest cases. The unconfounded delay oscillators identified in produced-capital systems (livestock and electricity-capacity cycles) do not contain the autonomously regenerating stock the resource model assumes. The per-system calculations use archived input series and are reported at that status in the Supplementary material (S4); three groups illustrate the screening logic.

Sheridan-6 groundwater (hydrological driver without a feedback rule) and Icelandic cod and haddock under harvest-control rules (rule present, no review-cycle identification) illustrate the confound patterns at case level (Supplementary S4). The four-state prediction is the stage-structured review map's closed loop of four state components (adults A, juveniles J, memory signal Z, held effort E), whose slow-stock class carries the centuries-scale dominant timescales of Section 3.3.

Peruvian anchoveta provides a further discriminator: a robust 3.7 yr catch periodicity with ENSO as candidate driver, evidence of shared periodicity rather than an identified mechanism (Supplementary S4 for the battery, multiplicity family, and confirmatory tests). The subannual review regime lies below the anchovy-class review-interval response region, so the unclassified controller prevents that comparison from testing the mechanism. The 3.7 yr catch periodicity is not a review interval.

Two structural hypotheses may help explain the zero count. The long records needed to test the claim exist primarily for visible crises, and visibility is correlated with a fast non-institutional driver. And on the continuous-delay parameterisation that produces the instability window ($rg \approx 1.5$–1.6) the predicted period is centuries — longer than any institution has held a fixed response rule — while the 3–12 yr windows belong to the stage-structured map. The two period scales are properties of different operators and should not be read as one prediction. The visible-record pattern — long series repeatedly associated with climate variability, cohort effects, infrastructure changes, or emergency interventions — suggests, without establishing, a selection mechanism in which systems receive intensive monitoring after complex crises; no causal identification of selection bias is claimed.

One documented system sits at the window's edge rather than inside it: the spruce budworm 30–40 yr outbreak cycle is endogenous predator–prey outbreak dynamics rather than institutionally driven (Supplementary S4 for the rate mapping and window comparison). The system corroborates the mechanism class, not the institutional claim.

3.8 The northern cod two-window split

The case's positive content is a descriptive partition, offered as a consistency check of the assessment-window mechanics rather than an empirical rejection of scalar autonomous or single-driver dynamics. **The data split the phenomenon into two events — the crash interpretation is formulation-dependent, and the post-collapse dynamics expose a second identification problem not resolved by the mortality-allocation comparison. The split is the positive content, not a new mechanism.**

The crash window (1991–1995) is documented by the assessment table (DFO CSAS SAR 2016/026, Table A2; Table 2): spawning stock biomass falls from 735 to 10 kt across the five calendar years 1991–1995 while estimated natural mortality reaches 2.2–2.6 yr^{-1}, roughly ten times pre-collapse levels (the pre-collapse reference period, estimator, and uncertainty interval for this ratio are documented with the assessment table; Table 2 begins in 1991). The displayed values are rounded values from the assessment table . The interpretation of the crash is a formulation attribution, not a causal claim. The NCAM M-shift formulation allocates most estimated mortality to natural death, and NCAM's M is an estimated unobserved-death component conditional on model structure — assessment-framework proceedings explicitly caution that unreported fishing deaths may enter de-facto M (Cadigan, 2016). The constrained-M formulation attributes the crash to unreported catch. The constrained-M quantities are unreproduced (Supplementary S13.2 for the quantities and reproduction requirements): they are listed as open problems and stated here as hypotheses, not results.

Table 2. Northern cod spawning stock biomass (SSB) and estimated instantaneous natural mortality (M) from DFO CSAS SAR 2016/026 (Table A2, NCAM M-shift formulation), rounded from the assessment table. The survival column $\exp(-M)$ is a transformation of the reported M — annual survival conditional on the estimated natural-mortality component alone, since fishing mortality and other modelled removals also apply — not an independently observed survival series.

Year	SSB (kt)	M (yr^{-1})	$\exp(-M)$
1991	735	1.002	0.367
1992	382	2.214	0.109
1993	101	2.575	0.076
1994	31	2.331	0.097
1995	10	0.288	0.750
1996	16.05	0.341	0.711
2000	34.42	0.717	0.488
2004	20.07	0.362	0.696
2005	25.18	0.288	0.750
2010	96.91	0.696	0.499
2015	298.65	0.278	0.757

The non-recovery window is the second event. After the moratorium, the series both rises and falls across tens of thousands of tonnes (Table 2; the full annual record with uncertainty intervals is the assessment table). The phase-line obstruction of Section 2.7 applies as a model-class diagnostic (not an empirical rejection of scalar autonomous models for northern cod): an error-free trajectory with repeated direction reversals cannot arise from a fixed scalar autonomous model with constant removals. The assessment record does not satisfy the diagnostic's two conditions — the SSB series is an assessment estimate rather than an exact trajectory, and removals were not fixed through the window — so the proposition is a statement about model adequacy for this class, not a direct empirical rejection. Two scope statements govern the result. The reliable contradiction is repeated direction reversal: convergence toward K can be arbitrarily slow near degeneracy, so failure to reach an unspecified biomass by an unspecified deadline is not a contradiction. And the obstruction must be separated from rejection under measurement error, process noise, age structure, migration, time-varying mortality, and state-space observation models — the incompatibility is a property of the exact trajectory class, nothing broader.

The subsequent record does not close the second window. Rose (2026), comparing two surplus-production reconstructions spanning 1983–2023 (Rose and Walters, 2019; DFO, 2024), documents that surplus production and stock growth stalled after 2015, with some years negative, and that the stall-point biomass remains controversial. Structural-equation analyses in the same review attribute the production deficit to capelin limitation and harp seal predation rather than fishing alone. For the present analysis the key point is the persistence of the non-recovery:

a decade after the rebound reported by Rose and Rowe (2015), the post-2015 production stall is inconsistent with the simplest fixed-parameter surplus-production interpretation considered here, which is precisely what the split's second window — a second identification problem not resolved by the mortality-allocation comparison — states. Read through Lemma 2.2(ii), the seal-predation attribution carries a directional consequence: to the extent predation acts as additional mortality on the stock, the effective threshold facing recovery lies above $\mathfrak{s}$, so predation raises the threshold for re-establishment rather than merely slowing convergence. No predation-mortality rate is estimated here; the lemma is used for the direction of the effect, not its magnitude.

The ecosystem context (Supplementary S5) is background only: the harp seal increase and capelin decline in the mass-balance record (Tam and Bundy, 2019) are descriptive inputs, not causal tests.

Figure 5 plots the Table-2 SSB and mortality series against the two windows.

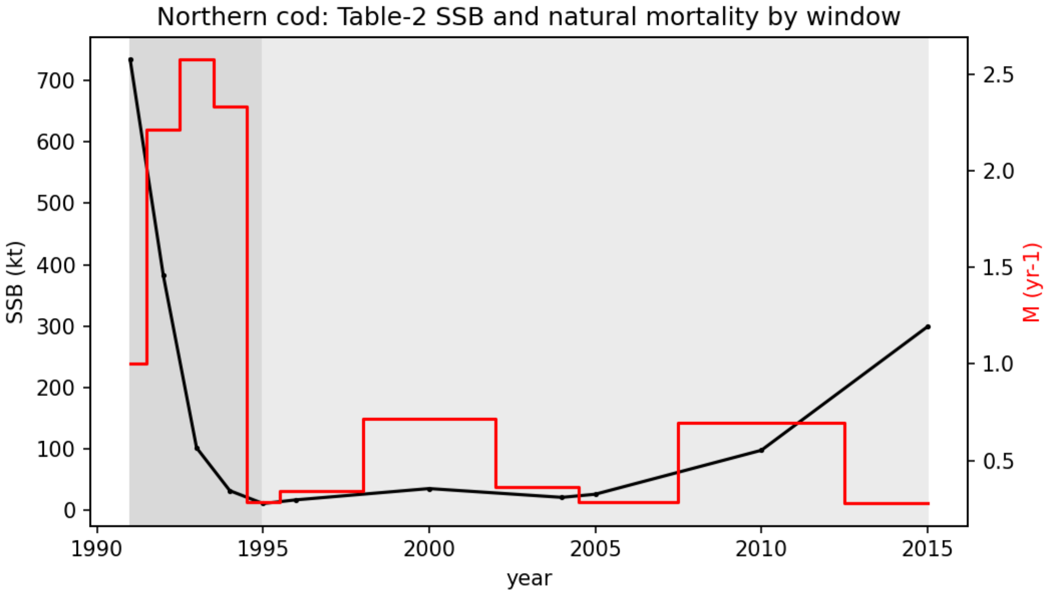

Figure 5: Northern cod Table-2 series by window. Black: spawning stock biomass; red steps: instantaneous natural mortality; shaded: the crash (1991–1995) and non-recovery (1995–2015) windows.

Table 5 collects the Section 3 results with their evidential status; Supplementary S1 is the complete inventory.

Table 5. Results at a glance: each Section 3 result with its evidential status.

Forward invariance (Section 3.1)	theorem
Rapid-review consistency (Section 3.2)	conditional theorem (finite horizons)
Response regions (Section 3.3)	archived records (provisional)
Crossing record; battery, F-grid, NS records (Section 3.4)	re-execution-verified computation; nominal-tier records (S11)
Screen null; alternative bands; F-excess; ADH (Section 3.5)	re-execution-verified zero count; nominal-tier records (S9.1, S11, S12)
Power analysis (Section 3.6)	independently re-executed computation
Case search (Section 3.7)	executed search (evidential gap, not disconfirmation)
Cod two-window split (Section 3.8)	descriptive partition; hypotheses, not results

4 Discussion

4.1 Architecture substitution: what changes when the operator changes

The operator-spectra results of Section 3.4 are the paper's core methodological finding. Stability windows are operator-specific: they do not transfer across review-map operators. The same feedback loop — surplus production, an institutional deficit signal, an effort law — carries an archived, unreproduced instability record near 3–4 yr of review under the stage-structured map, convergence over a 1–20 yr grid under that map's cod-class parameterisation, and instability at annual review with a complex crossing near 6.5 yr under the exact logistic hold map (full records, the Euler command-step artefact, and the q-sensitivity caveat in Section 3.4 and its Reading; the plant–operator confound is stated alongside and the operator effect is not claimed isolated).

The complete crossing record sharpens the operator finding: on one plant the exact and Euler updates disagree on the number and location of crossings (Section 3.4). The review cadence is a local spectral design parameter, and the command step is a second, independent one. None of these statements can be obtained from the continuous-delay equation, and none transfers to it. Any empirical or theoretical claim about institutional feedback must therefore state its operator: a stability window located under continuous delay is not a prediction about an institution that reviews periodically, and an annual-review result does not generalise to multi-year plans. The governance architecture itself is a testable component of the hypothesis, not a re-parameterisation detail.

4.2 What the null does and does not establish

The 42-stock screen returns a spectral null, and its value lies in what it refutes: the claim, implicit in treating periodicity as evidence, that robust target-band cycles are common in assessed-stock records. It does not establish the contrary. With anchovy-class power between 0.02 and 0.14 across tested noise and record-length regimes, the screen cannot adjudicate the extractive mechanism for that class; the sprat-class result is informative only in favourable regimes; and no screen of retrospective series can compare controller signs, because the sign is not in the data. This is the same null discipline that the skeptical re-analysis of the Russell Cycle applied to climate-forced cycles (McManus, Licandro, and Coombs, 2016), transferred to governance-feedback claims: a diagnostic is not a causal claim, and no accumulation of diagnostics converts to causal evidence.

4.3 The cod case at its exact status

The cod case contributes a descriptive partition, not a mechanism: crash-window mortality is formulation-dependent (Section 3.8), and the post-collapse record is not resolved by either mortality-allocation formulation, with the post-2015 production stall persisting under independent reconstructions (Rose, 2026). The obstruction mathematics applies within the scope stated in Sections 3.8 and 4.7(vii). What the case supplies is a falsification benchmark: a well-documented collapse-and-stall record against which prospective designs can be powered and scored.

4.4 A falsification standard for institutional-feedback claims

The model family gains empirical content by specifying the outcomes that count against its mechanism and parameterisation. Five outcomes:

1. If independently measured implementation and deployment lags do not covary with the instability bands predicted under fitted resource parameters, the quantitative delay claim is weakened.

2. If an observed oscillation remains after climate, cohort, and regime forcing are modelled, but its phase relation between decline signal, effort, and biomass contradicts the controller's causal ordering, the mechanism is rejected for that case.
3. If sampled-data analysis removes a continuous-delay band under realistic review rules, the continuous model cannot be used for that institution.
4. If a protective controller dominates the extractive controller across uncertainty without inducing the claimed volatility, no anti-regulation conclusion may be drawn.
5. If realistic record lengths have low power for the predicted signal, field spectral nulls cannot adjudicate the mechanism; prospective or experimental evidence is required. The Icelandic cod case illustrates the bind: its post-1995 variability (coefficient of variation 0.37–0.39; no dominant spectral peak, Supplementary S11) overlaps the continuous-delay band (9.9–20.3 yr) in timescale yet carries no institutional signature, because a competing environmental driver is empirically supported for the same outcome (criterion (iii); Supplementary S4).

4.5 Prospective designs

The retrospective evidence does not identify the institutional mechanism; the constructive content of this paper is the programme that could. Retrospective evidence can still discipline the mechanism where a prespecified contrast exists: the T_r-ranked cross-sectional test across the 42-stock cohort is executed in Supplementary S10 and returns a controlled null — no window-approach gradient, with $T_r \approx 1$ dormancy holding at 0/42 and 1/42 flags in the two bands. Five designs are specified as preregistration targets; none has been executed, and each is intended for preregistration before outcome inspection — no registration identifier or archived protocol exists yet, and none is claimed.

Governance-event panels. For each resource–jurisdiction unit, a source-linked event record dates each stage from raw observation to ecological response, with date uncertainty kept distinct rather than collapsed to one lag (full specification in Supplementary S13.3). The cod case supplies two dated decision events — the moratorium announcement of 2 July 1992 and the reopening of 26 June 2024 with an 18 kt total allowable catch — with two negative constraints: no governance lead is inferred from annual biomass data, and a fast response does not establish adequacy.

Quasi-experimental timing. Event-study, difference-in-differences, synthetic-control, interrupted-time-series, or state-space intervention designs on schedule variation plausibly independent of the outcome innovation, with preregistered estimands (full specification in Supplementary S13.3). Two coding rules are binding: a change in implementation lag is not coded as a change in T_r, and a nominal schedule change is not a treatment unless it changes a responsive decision opportunity.

Out-of-sample mechanism comparison and the displacement discipline. Held-out comparison across environmental-forcing, cohort-resonance, institutional-delay (specified controller sign), combined, and null time-series models, with frozen splits and declared outcomes (full specification in Supplementary S13.3). The displacement rule: a delay explanation is weakened when it cannot reproduce the pre-registered phase ordering, and displaced when a competitor predicts the held-out observations better.

Controlled and randomized human-in-the-loop experiments. Randomised review cadence and feedback timing or sign in a common simulated renewable-resource environment, with the gain–phase signature as the diagnostic target (full specification in Supplementary S13.3). The randomisation described is a laboratory or advisory-interface design, not a field intervention on a live stock; mechanism-class precedent exists (Costantino, Cushing, Dennis, and Desharnais, 1995; Moxnes, 1998) but is not validation of the institutional equations.

Closed-loop management strategy evaluation. Policy comparison in a closed loop keeping six uncertainty classes distinct (the management-procedure tradition: Punt and Donovan, 2007), crossing review timing, delays, controller sign, observation error, parameter draws, operating-model class, and process-noise regime (full specification in Supplementary S13.3). At minimum, responsive extractive, responsive protective, and fixed-plan controllers are compared; a conclusion that compares only two review intervals inside the extractive class cannot be generalised to governance as a whole. Three wider empirical hypotheses of the programme's general theory — observation aggregation, governance phase ordering, and substitution certificate — are adopted here only as motivation, and otherwise explicitly deferred: governance phase ordering supplies the gain–phase diagnostic target the designs above inherit, while testing any of the three requires prospective, adequately powered designs this paper specifies but does not execute.

4.6 Distributive constraints where reproducible

The social side of the cod case is presented at measurement level (Appendix B): the constituency-definition mismatch table (licence holders against census-subdivision residents; fishing income against all-resident sector income; the unspecified floor), the Statistics Canada community-level series (tables 38-10-0167-01 and 38-10-0168-01), the measured-floor construction (I_k, c_k) with its componentwise non-decline rule, and the admitted world-hook instruments are given in Appendix B, with the full instrument detail and the unreproduced pipeline register in the Supplementary material (S6). What the case does not measure is the point: no licence-holder panel exists, the community series is not one, and no measured floor is operational — the distributive constraints remain stated rather than resolved (Section 4.7 (viii)).

4.7 Limitations

(i) Diagnostics are not causal claims: the spectral null, the response regions, the power values, the case calculations, and the cod split carry their stated types and no more.

(ii) The stage-structured response regions are exploratory finite-grid, trajectory-classified records pending complete stage registration and multiplier analysis; the logistic hold-map multiplier record is complete (Section 3.4), and the stage operator's multiplier and trajectory record is supplied by the reconstruction of Section 3.4 — a newly specified object whose comparison with the archived windows is a consistency check, not a validation. No Poincaré-map multiplier classification is claimed for the archived stage-output values, and they are not reproducible numerical propositions until the original computational record is attached. The effort scale is uncalibrated: results are reported at imported catchability values, and reparameterising the controller in fishing mortality ($F = qE$) is the stated next step for the stage operator.

(iii) The two review-map operators are distinct: statements computed on the hold map, on the stage-structured map, and on the continuous-delay equation do not transfer to one another.

(iv) The screen is a selected-cohort consistency check whose power is high only in favourable noise and record-length regimes; the null is not proof of absence and adjudicates nothing about controller sign; the AR(1) null may understate low-frequency power. The eleven-variant battery of Supplementary S9.1 holds the zero count under ARMA(1,1), trend-stationary, block-bootstrap, detrending, and regime-shift nulls; the remaining masked-peak risk runs via overestimated null power (unmodelled observation-error or retrospective-bias variance raising the threshold), not the underestimation above.

(v) The structured case search does not independently disconfirm the mechanism: finding no eligible cases under these criteria is not evidence against it.

(vi) The cod case establishes a descriptive partition, and no mechanism is identified.

(vii) The obstruction mathematics concerns exact trajectories of the fixed-parameter, fixed-removals autonomous class and is not a rejection under measurement error, process noise, age structure, migration, time-varying mortality, or state-space observation models.

(viii) The social object is not operationalised: the human series is not assembled, the normative floors are unoperationalised, and documenting the gap is the finding.
(ix) The prospective designs are preregistration targets, not executed studies, and they do not retroactively change the status of the retrospective results.

4.8 Design principles for the governance clock

The results support four design principles for the decision clock (Box 2).

Box 2. Design principles for the governance clock.

Principle	Content
Treat timing as a control variable	The review interval is optimised alongside harvest limits in management strategy evaluation; annual review is not assumed stabilising.
Match the rule to the clock	Protective rules tolerate institutional rigidity; extractive rules require frequent review to contain destabilisation.
Decouple observation from decision	High-frequency monitoring does not require high-frequency revision; multi-year plans are stable under protective rules.
Design for structural breaks	Scheduled review is paired with asynchronous emergency triggers (circuit breakers) that bypass the review clock during regime shifts.

Implications for management design. Eight implications follow for review-interval choice.

(1) The review interval is a control variable: stability is computed per interval, not per loop (Section 3.4).
(2) Annual review is not automatically stabilising: the baseline annual verdict is unstable with a near-unit-circle margin (Section 3.4).
(3) Protective rules are more robust across review intervals than extractive rules: the exact protective map is stable at every tested interval (Section 3.4).
(4) Fixed and responsive plans are not equivalent instruments: a held multi-year command and an annually revised rule are different operators (Sections 2.6 and 4.4).
(5) Emergency triggers are modelled separately from scheduled reviews: structural breaks between reviews are invisible to the fixed clock (Sections 3.8 and 4.5).
(6) Management strategy evaluation treats the review interval as an explicit factor alongside the control rule (Section 4.5).
(7) Spectral periodicity is not evidence of institutional cycles: the screen's null carries no controller-sign information (Sections 3.5–3.7).
(8) Observation, assessment, decision, deployment, and response lags are not collapsed into one governance lag (Section 2.1).

Beyond fisheries, the framework is reusable as a modelling discipline: it answers how periodic governance should be represented in ecological models, how the stages of the advice chain should be kept distinct, and how modellers can test whether a stability claim survives a change of decision operator.

The logistic core is illustrative; the structural results are the operator-dependence of the stability verdict, the finite-horizon scope of the rapid-review limit, and the non-equivalence of one-step and exact updates as a caution within the studied class.

Sample-and-hold architectures beyond fisheries face the same clock-mismatch failure mode: five-yearly climate pledges held between global stocktakes (UNFCCC, 2015) align a political review clock with ecological and environmental clocks only by design, not by default.

5 Conclusion

Periodic review is the operating architecture of fisheries governance, and it is not well represented by either of its standard formalisations. The sample-and-hold model specified here makes the governance clock an explicit object: seven time objects, a projected explicit controller, and a spectrum computed per operator rather than per loop. Its empirical layer contains one case whose positive content is a descriptive split — the northern cod two-window partition — and a falsification discipline for the rest: a selected-cohort screen with no target-band discoveries and stated power, a zero-count case search with its structural hypotheses, and a falsification standard with five outcomes that count against the mechanism. The mechanism is falsifiable, and the five prospective designs specify how it would be falsified, comparatively supported, or left unresolved. Until those designs are executed, the empirical layer stands as stated here: a well-posed mechanism, an exploratory computational record, limited screen power, a case search that does not disconfirm, and one descriptive split. The decision clock — the timing and form of policy revision — is therefore a design variable with stability consequences.

Appendix A. Registration and reproducibility record (consolidated)

This appendix collects the registration and reproducibility requirements stated in Sections 2.2, 2.4, 2.5, 3.3, and 3.4. The convention: an analysis is pre-registered when its plan is dated and fixed before any run, as the stage-map reconstruction of Section 3.4 is; computational artifacts (solver configuration, seeds, identifiers, calibration records) are archived with the materials. The consolidated requirements:

- The solver configuration and initial histories are registration requirements; until that computational record is complete, the stage-output values carry provisional status (Sections 2.2 and 3.3).
- The RAM stock identifiers and the eligibility table are archived with the computational materials (Section 2.4).
- The full null-calibration record — AR(1) coefficient estimation, detrending inside each null replicate, missing-data treatment, and the number of Monte Carlo replicates — is archived with the computational materials (Section 2.4).
- The simulation code and seeds are archived with the computational materials (Section 2.5).

Two further consolidated records. First, the logistic-core parameter vector (r, K, $E_{\max}$, δ_0, Z_{ref}, Δ_{ref}, τ_m; $\delta = \log 2/10$ is stated in Section 2.1) and the linearised fixed point (N^*, E^*, Z^*) with its interiority to the nonsmooth regions of Φ_k and $\Pi_{[0,E_{\max}]}$ are printed in Table 3, with the code constants block pinned per row; the monodromy's numerical construction and the continuous eigenvalue λ and crossing angle θ of Section 3.4's records remain part of the computational record. Second, the Data availability statement records the deposit status of the same requirements (including the reconstruction's dated plan, campaign code, and output tables); the prospective designs of Section 4.5 remain preregistration targets with no registration identifier or archived protocol. The full discharge register is Supplementary S8.

Table 3. Parameter values for the logistic hold-map core and the stage-structured reconstruction. No value here is newly computed except the linearised-gains, crossing-angle, eigenvalue, and solver rows, which are computed in Supplementary S11; the baseline-core vector is printed here, with the code constants block pinned per row, and values not stated remain in the computational record (this appendix). Time is in years; biomass, effort, and signal scales are bare model units (K, $E_{\max}$, Z_{ref}, and Δ_{ref} set the scales).

Object	Parameter	Value as stated	Stated at
Logistic hold-map core	catchability q	0.001	Section 3.4 (stage Plant paragraph, as the hold-map core's value)
Logistic hold-map core	softplus sharpness k	10	Section 3.4 (Plant paragraph)
Logistic hold-map core	effort-response coefficient η	0.914	This table; constants block `droop_test.py` L49–58 @ 24c980cd; inherited illustrative baseline (Candidate-A value from the companion delay study, whose institutional coefficients are explicitly uncalibrated: Abaee, 2026); the delay-Hopf window opens between $\eta = 0.5$ and 0.7 and the $r = 0.02$ inclusion threshold sits at $\eta^* \approx 0.85$, so 0.914 is neither threshold (basis runs deposited; Supplementary S9.2)
Logistic hold-map core	growth rate r (yr^{-1})	0.02 (validation argument)	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	carrying capacity K	100	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	effort ceiling $E_{\max}$	30	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	effort-law gain δ_0	0.01	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	reference deficit Z_{ref}	1	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	reference scale Δ_{ref}	1	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	memory timescale τ_m (yr)	5	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	memory shift δ	$\log 2/10 \approx 0.069$ (equilibrium memory level)	Section 2.1
Logistic hold-map core	linearised fixed point (N^*, E^*, Z^*); interiority	$(89.55188, \delta, 2.08962)$; $E^* \in (0, E_{\max})$, $\Phi_k(0) = \delta$, $sp_k'(0) = 1/2$	This table; equilibrium routine `base_equilibrium`, `droop_test.py` L86–93 @ 24c980cd; closed form in Supplementary S2.3
Logistic hold-map core	crossing record	47.536, 79.143, 2.306, 6.501 yr; $\rho = 1.00035$, 1.00055, 0.9838, 0.9967	Section 3.4

Object	Parameter	Value as stated	Stated at
Logistic hold-map core	scan construction	200,001-point scan with bisection over $[0.2, 200]$ yr	Section 3.4
Logistic hold-map core	linearised mobilising gains (C_E, C_Z)	$(-0.059518, +1.785019)$	Supplementary S11 (sensitivity battery)
Logistic hold-map core	crossing angle θ_0 (exact map, 6.501 yr)	0.175494 rad	Supplementary S11
Logistic hold-map core	continuous eigenvalues λ (undelayed loop)	-0.278147, $+0.000360 \pm 0.027683i$	Supplementary S11
Logistic hold-map core	derivative construction	closed-form monodromy (no finite differences)	Section 3.4
Logistic hold-map core	solver; precision; grid check	matrix exponential (`expm`); IEEE-754 double; 2k/20k/200k scans identical; 50-digit certificate (S12)	Supplementary S11
Stage reconstruction	class natural mortality M and recruitment delay τ	anchovy (0.90, 1), sprat (0.40, 2), cod (0.20, 5), slow-stock (0.045, 25) yr	Section 3.4 (Plant paragraph, with sources)
Stage reconstruction	steepness h	0.75; sensitivity layer $\{0.6, 0.9\}$	Section 3.4
Stage reconstruction	scale and survival	$A_0 = 100$; $s_A = s_J = e^{-M}$; $\beta = (c-1)/A_0$, $\alpha = c(1-s_A)/s_J$, $c = 4h/(1-h)$	Section 3.4
Stage reconstruction	catchability q	0.001; sensitivity 0.1	Section 3.4
Stage reconstruction	softplus sharpness k	10	Section 3.4
Stage reconstruction	derivative construction	central finite differences, step 10^{-6}, chain-rule cross-check to 10^{-4}	Section 3.4
Stage reconstruction	scan grid	$T_r \in \{1, \ldots, 50\}$ yr at annual internal steps	Section 3.4
Stage reconstruction	trajectory classification	2000 review-steps, tail 500, relative tail standard-deviation thresholds 2% and 0.1%; initial condition at the unfished plant equilibrium, memory filled, effort at half the equilibrium	Section 3.4

Table 4. Notation: every symbol with its definition site. No row introduces a symbol; rows without a site would be gaps, and all rows resolve.

Symbol	Meaning	Defined at
N	resource stock	Section 2.1
Z	institutional deficit signal	Section 2.1
E	extraction effort	Section 2.1
$\mathcal{P}_{T_r}$	review Poincaré map (sampled flow at review instants)	Section 2.1
$D\mathcal{P}_{T_r}(X^*)$	monodromy matrix (Jacobian at the fixed point)	Section 2.1
Π	projection onto $[0, E_{\max}]$	Section 2.1

Symbol	Meaning	Defined at
Φ, Φ_k	signal map; softplus realisation (shift δ, sharpness k)	Section 2.1
δ; δ_0	equilibrium memory level $\log 2/10$; effort-law gain (unrelated)	Section 2.1
F_B; F_B^{prot}	extractive effort law; protective comparator	Section 2.1
$\mathcal{A}_n$; $\mathcal{O}$	assessment operator; observation operator	Section 2.1
$S(\cdot)$; S	surplus production (control sections); spawning stock biomass (cod sections, Table 2)	Section 2.1; Sections 2.7, 3.8
g	maturation delay (delayed-recruitment parameterisation)	Section 3.3
C_E, C_Z	linearised effort-law coefficients (per channel; protective values in Section 2.1)	Section 3.4 (Abaee, 2026, equation (3))
A, J, Z, E	four-state loop: adults, juveniles, memory signal, held effort	Section 3.7
$\mathfrak{s}$, K, $C(t)$, M, M_x, F	cod case: unstable threshold, unexploited capacity, removals, natural mortality, extra mortality, fishing mortality	Sections 2.7, 3.8; Table 2

Appendix B. Distributive constraints where reproducible

Where the social side of the cod case can be presented, it is presented at measurement level. The worst-off relevant population is a modelling choice — a named constituency — and the candidate constituency for 2J3KL is the registered inshore harvesters and licence holders. The measurement record against that candidate is the following mismatch table, each row stated rather than resolved:

Object	Candidate definition	Available measure	Mismatch	Required data
Population	registered licence holders	census-subdivision residents	not the same population	licence microdata
Income	fishing income	all-resident sector income (incl. aquaculture, processing)	includes non-fishing income	administrative income
Floor	specified cutoff (I_k, c_k)	none operational	not specified	specified instrument and cutoff

The available community-level series (Statistics Canada tables 38-10-0167-01 and 38-10-0168-01; mean income from fishing falling from 32.2% to 25.6% across 43 fishing-dependent census subdivisions, 2016–2021) is all-resident employment income including aquaculture and processing. The constituency must therefore either be redefined to the census-subdivision populations, or licence-holder microdata or administrative data must be obtained — the community table is not a licence-holder panel. A measured floor is a pair (I_k, c_k) — an instrument and a cutoff — a measurement object distinct from the norm that motivates it, and the componentwise rule is that the arrangement fails as soon as any measured margin $m_k(G,t) = I_k(G,t) - c_k$ is negative; the conjunction admits no compensatory master scalar. Non-decline is a normative rule, not automatically an empirically justified floor: baseline, cohort, inflation and purchasing-power treatment, uncertainty, attrition, acceptable variation, authority, and structural diversification all must be specified before a floor measures anything. The instruments admitted as world-hooks (the global Multidimensional Poverty Index of Alkire and Foster, 2011; the World Bank Poverty and Inequality Platform) carry release and vintage locks, and what they cannot capture

— who pays for persistence, who may use remaining variety, whose knowledge counts, future persons — is stated as a boundary and left open rather than filled with a scalar score. The full instrument detail and the unreproduced pipeline register are in the Supplementary material (S6).

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Supplementary material is deposited with this article (the accompanying Supplementary file): the statement inventory with the status of every main-text statement (S1), the remaining proofs and identification results, including the dimensionless identifiability chart (S2), the epistemic-layer results (S3), the case-screening records at full detail (S4), the ecosystem context and toy tests (S5), the distributive-layer detail (S6), the empirical hypotheses with specified tests (S7), the reproducibility register (S8), and the screen-sensitivity battery and stage-scan decomposition records (S9), and the T_r-ranked test record (S10). The register itself accompanies the article; the computational artifacts it inventories (stage registration, solver configuration, query log, nonlinear exact-update comparison) are registration requirements tracked in the register. The stage operator's boundary record is supplied by the pre-registered reconstruction of Section 3.4 — a newly specified object with archived plan, code, and outputs — while the archived stage-output values keep their provisional status until the original computational record is attached.

Declarations

Data availability

The stock and assessment series analysed in the spectral screen are drawn from the RAM Legacy Stock Assessment Database v4.66 (public release). The northern cod assessment values are published in DFO CSAS SAR 2016/026. The computational record of the response-region analyses (solver configuration, initial histories, stage registration), the nonlinear exact-update comparison of Section 3.4 are registration requirements; the case-screening table and query log are deposited with the article; the corresponding stage-output values carry provisional status until those artifacts are attached. The analysis datasets (RAM stock identifiers and eligibility table; processed spectral series; case-screening table and query log; T_r-ranked test table; ICES cod and haddock series; η-basis window runs) and the analysis code (spectral-screen routines; power-simulation code and seeds; sensitivity-battery code and logs; case-screen, T_r-ranked test, Icelandic-cod audit, and η-basis scripts with their logs) are deposited with the article. The stage-map reconstruction of Section 3.4 is fully specified and will be deposited with the article: the dated preregistration plan (parameter table, comparison criteria), the campaign code, and the five output tables (equilibria, multiplier record, trajectory classification, robustness experiment, steepness sensitivity). Community-level social values are drawn from Statistics Canada tables 38-10-0167-01 and 38-10-0168-01 with the population-mismatch caveat stated in the text.

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The author declares no competing interests.

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